INCH-POUND

MIL-P-70565B (AR)
15 September 1993
SUPERSEDING
(See 6.7)

MILITARY SPECIFICATION

PROCESSOR, FIRE CONTROL: 9360890

This specification is approved for use by the U.S. Army Armament, Munitions and Chemical Command, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers the requirements, examinations and tests for the Fire Control Processor (FCP) used in Gun, Anti-Aircraft, Artillery, Self-Propelled 20MM, M163A2, and Towed 20MM, M167A2. The FCP uses two 16-bit microprocessors interconnected into a master-slave arrangement that are controlled by an interrupt driven executive and three subprograms: fire control, built in test (BIT) and servo control. The executive and the first two subprograms are resident on and executed by the computer memory board, and the third subprogram is resident on and executed by the Servo Controller Board (SCB). Communication between the two processors is provided by a common memory block located on the SCB. The servo control subprogram operates autonomously during degraded system operations (i.e., in ground mode). The FCP receives various analog, discreet, digital (RS-422 serial transmission) and synchro signals from several system components, including operator commands, and processes these signals to generate the various control signals required for system operation. The FCP provides sight stabilization, turret commands, and gun lead angle, and provides regenerative tracking assistance. The BIT subprogram provides a self test and numerous system tests to check out operational status and to

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document, should be addressed to: Commander, U.S. Army ARDEC, ATTN: SMCAR-BAC-S, Picatinny Arsenal, New Jersey 07806-5000 by using the Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

AMSC N/A **FSC** 1285

DISTRIBUTION STATEMENT A. Approved for public release; distribution is unlimited.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

fault isolate malfunctions. Signal scaling and conversion is performed within the FCP. Full description of the software and system operation is provided by the Computer Program Performance Specification (SW 12563296). The FCP contains five circuit card assemblies, a mother board and a cable assembly (W8) (see 6.1).

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications, standards, and handbooks. The following specifications, standards, and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the issue of the Department of Defense Index of Specifications and Standards (DODISS) and supplement thereto, cited in the solicitation (see 6.2).

SPECIFICATIONS

MILITARY

MIL-F-13926	- Fire Control Materiel, Manufacture and Inspection, General Specification for
MIL-P-14232	- Parts, Equipment and Tools for Army Materiel, Packaging of
MIL-I-45607	- Inspection Equipment, Acquisition, Maintenance and Disposition of
MIL-C-70567	- Circuit Board, Computer and Memory (PIVADS): 9360878
MIL-C-70568	- Circuit Board, Input/Output (PIVADS): 9360881
MIL-C-70569	- Circuit Board, Synchro Interface (PIVADS): 9360871
MIL-C-70570	- Circuit Board, Servo Control (PIVADS): 9360875
MIL-C-70585	- Circuit Board, Data Transmission (PIVADS): 9397510
MIL-C-70701	- Cable Assembly W8 (PIVADS): 9397464

STANDARDS

MILITARY

MIL-STD-109	- Quality Assurance Terms and Definitions
MIL-STD-810	- Environmental Test Methods and Engineering Guidelines

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the current version before use.

(Unless otherwise indicated, copies of federal and military specifications, standards, and handbooks are available from the DODSSP - Customer Service, Standardization Documents Order Desk, Bldg. 4D, 700 Robbins Avenue, Philadelphia, PA 19111-5094.)

2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this document to the extent specified herein. Unless otherwise specified, the issues are those cited in the solicitation.

DRAWINGS (see 6.4)

U.S. ARMY ARMAMENT RESEARCH, DEVELOPMENT AND ENGINEERING CENTER (ARDEC)

9360874	-	Backplane Schematic
9360890	-	Fire Control Processor Assembly
9360956	-	Fire Control Processor Schematic
9360957	-	Wiring Diagram, Fire Control Processor
9360965	-	Fire Control Processor Interface Device
9397458	-	Cable Assembly W1, Fire Control Processor
9397459	-	Cable Assembly W2, Fire Control Processor
9397460	-	Cable Assembly W3, Fire Control Processor
9397461	-	Cable Assembly W4, Fire Control Processor
9397462	-	Cable Assembly W6, Fire Control Processor
9397463	-	Cable Assembly W7, Fire Control Processor
9397464	-	Cable Assembly W8, Fire Control Processor
10545465	-	Isolator
12563264	-	Test Cable W1
12563265	-	Test Cable W2
12563266	-	Test Cable W3
12563267	-	Test Cable W4
12563268	-	Test Cable W5
12563269	-	Test Cable W6
12563270	-	Test Cable W7
12563271	-	Test Cable W8
12563272	-	Test Cable W9
12950877	-	Requirements for Soldering Electrical and Electronic Assemblies

SPECIAL PACKAGING INSTRUCTION

PS 9360890	-	Packaging Data Sheet, Fire Control Processor

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the currentversion before use.

OTHER GOVERNMENT DOCUMENTS

TPS12554988	-	Test Program Set, Fire Control Processor Assembly A21
SW12563296	-	Computer Program Performance Specification, Product Improved Vulcan Air Defense System (PIVADS)
SW12563297	-	Interface Design Specification, M163A2 and M167A2, Product Improved Vulcan Air Defense
SW12563298	-	Program Design Specification
SW12563299	-	Program Description Document
SW12563301	-	Program Package Document

(Copies of other Government documents, drawings, and publications required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting activity.)

2.2 Non-Government publications. The following document(s) form a part of this document to the extent specified herein. Unless otherwise specified, the issues of the documents which are DoD adopted are those listed in the issue of the DODISS cited in the solicitation. Unless otherwise specified, the issues of documents not listed in the DODISS are the issues of the documents cited in the solicitation (see 6.2).

INSTITUTE FOR INTERCONNECTING AND PACKAGING ELECTRONIC CIRCUITS

EIA RS-422	-	Electrical Characteristics of Balanced Voltage Digital Interface

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, New York 10018-3308.)

2.3 Order of precedence. In the event of a conflict between the text of this document and the references cited herein, the text of this document takes precedence. Nothing in this document, however, supersedes applicable laws and regulations unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Materials. Materials shall be in accordance with the applicable drawings and specifications.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the currentversion before use.

3.2 Components and assemblies. The components and assemblies shall comply with all requirements specified on Drawing 9360890, all associated drawings, and with all requirements specified in applicable specifications and standards. Where applicable, the FCP shall meet the requirements of MIL-F-13926 as follows:

a. Dimensions and tolerances.

b. Inorganic protective surface finishings.

c. Part identification and marking.

d. Workmanship.

3.2.1 Fabrication. The fire control processor shall be manufactured in accordance with Drawing 9360890 and drawings pertaining thereto. The subassemblies of the fire control processor shall have been inspected and accepted in accordance with their respective drawings and specifications, excluding environmental tests, prior to their installation into the fire control processor.

3.2.1.1 Computer and memory assembly. The computer and memory assembly shall be manufactured in accordance with MIL-C-70567.

3.2.1.2 Input/output assembly. The input/output assembly shall be manufactured in accordance with MIL-C-70568.

3.2.1.3 Synchro interface assembly. The synchro interface assembly shall be manufactured in accordance with MIL-C-70569.

3.2.1.4 Servo controller assembly. The servo controller assembly shall be manufactured in accordance with MIL-C-70570.

3.2.1.5 Data transmission assembly. The data transmission assembly shall be manufactured in accordance with MIL-C-70585.

3.2.1.6 Cable assembly W8. The cable assembly W8 shall be manufactured in accordance with MIL-C-70701.

3.2.1.7 Backplane assembly. The backplane assembly shall be manufactured in accordance with Drawing 9360959.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

3.3 Environmental service conditions.

3.3.1 Vibration. The FCP shall not suffer damage and shall meet the requirements of 3.5 after exposure to the equivalent of 6000 miles of track-laying vehicle operational vibration as defined by MIL-STD-810 narrowband random-on-random data for equipment installed on the deck of the M113 armored personnel carrier.

3.3.2 Shock. The FCP shall not suffer damage and shall meet the requirements of 3.5 after exposure to a total of 18 impulse shocks consisting of three shock impulses of 15 g's ± 10 percent, 11 milliseconds ± 10 percent, half-sine wave applied in each direction of three mutually perpendicular axes.

3.3.3 Altitude operating. The FCP shall meet the requirements of 3.5 during and after exposure to barometric pressure encountered at 10,000 feet above sea level (20.6 inches of Hg).

3.3.4 Temperature.

3.3.4.1 Operating. The FCP shall meet the requirements of 3.4 and 3.5 when stabilized thermally at any temperature between -40°F and +141°F.

3.3.4.2 Non-operating (storage). The FCP shall not suffer permanent damage when subjected to the constant temperature extremes of -60°F for a period of 24 hours (after temperature stabilization), and shall meet the requirements of 3.5 when returned to room ambient.

3.3.5 Humidity (non-operating). The FCP shall meet the requirements of 3.5 after exposure to an atmosphere with a relative humidity in excess of 95 percent for a period of 10 days.

3.3.6 Electromagnetic interference. The FCP shall be designed to be electromagnetically compatible with the M163A2/M167A2 Vulcan Air Defense System and installed communication equipment.

3.4 Electrical continuity. Prior to installation of the five circuit card assemblies, each conductor in the FCP shall be electrically continuous as specified by the wiring diagram (9360957), the FCP backplane schematic (9360874) and the seven cable assembly drawings (9397458-9397464).

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the current version before use.

3.5 Performance. A technical description of the FCP's software functions begins with the Computer Program Performance Specification (CPPS), SW12563296, and continues in greater detail with the Program Design Specification (PDS), SW12563298 (three volumes), and the Program Description Document (PDD), SW12563299 (three volumes), and culminates with the Program Package Document (PPD), SW12563301, which provides the source and object codes. The Interface Design Specification (IDS), SW12563297, provides explicit definitions for the numerous hardware/software variables.

3.5.1 Performance requirements.

3.5.1.1 Power input/output. The FCP shall operate when the voltages of Table I are applied to the connector-pins shown in the FCP INPUT column of Table I. The FCP shall output the same voltages at the connector-pins shown in the FCP OUTPUT column of Table I.

TABLE I. Power requirements.

VOLTAGE	CURRENT (POWER)	FREQUENCY	RIPPLE	FCP INPUT	FCP OUTPUT
115 ± 5.75 Vrms	50 VA at 0.7 power factor lagging)	400 ± 0.4 Hertz sine wave with 10% maximum distortion	N.A.	J2-M (HI) J2-K (LO)	J4-J (HI) J4-K (LO) J7-X (HI) J7-Y (LO)
+5 ± 0.25 Vdc	15 A Max.	dc	20 mV rms	J2-D,E J2-P,F (RTN)	J7-(K),(R) J7-(M),(S) (RTN) J3-GG,HH (RTN) J8-HH (RTN)
+15 ± 0.75 Vdc	1.6 A Max.	dc	20 mV rms	J2-A J2-N (RTN)	J1-(F) J1-U (RTN)
-15 ± 0.75 Vdc	1.6 A Max.	dc	20 mV rms	J2-B J2-N (RTN)	J1-M J1-U (RTN)
+24 ± 1.2 Vdc	1.0 A Avg. 2.0 A Peak	dc	20 mV rms	J2-H J2-R (RTN)	J4-(R) J4-(M) RTN
-24 ± 1.2 Vdc	1.0 A Avg. 2.0 A Peak	dc	20 mV rms	J2-J J2-R (RTN)	J4-(N) J4-(M) RTN
+24 (BAT) +6/-2 Vdc	3.1 A Max.	dc	0.5 V P-P	J1-(C),(R),W J1-R (RTN)	J4-R J6-(B),(R) RTN J3-X (RTN)

NOTE: 1. The table low-side (LO) input/outputs and returns (RTN), except the +24 V (BAT) returns, and the following SHIELD connector-pins shall be grounded to the FCP chassis; i.e., shall have 0.2 ohm maximum resistance from connector-pin to chassis.
J2-C,G,L,S J3-E,W,(N),(T),(X),(Z) J4-L,(A),(K)
J7-F,Z,(D),(E),(F),(G),(H),(N),(T) J8-C,F,L,W,(H),(Z)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

3.5.1.2 Fan. With +24 V (BAT) applied to the FCP per 3.5.1.1, the internal fan shall rotate at 6500 ± 3050 rpm to provide air circulation for cooling.

3.5.1.3 Serial data communication interfaces.

3.5.1.3.1 Data clock outputs. The following ports shall output 500.000 kHz ± 74 Hz differential RS-422 square waves. These square waves shall be used to clock communications on Universal Asynchronous Receiver-Transmitter links between the FCP and external serial binary interfaces. The corresponding data rate shall be one sixteenth of the clock frequency (i.e., 32.250 kHz ± 5 Hz).

a. CONTROL PANEL CLOCK AT J7-R (HI) and J7-S (LO)

b. DISTRIBUTION BOX CLOCK at J8-P (HI) and J8-R (LO)

c. UNIT 2 CLOCK at J7-(C) (HI) and J7-H (LO)

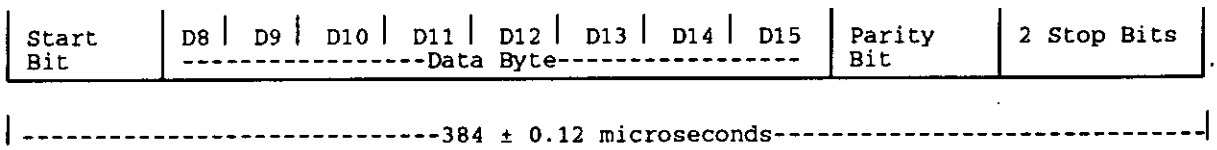
d. UNIT 4 CLOCK at J6-K (HI) and J6-J (LO)

e. UNIT 5 CLOCK at J6-B (HI) and J6-C (LO)

3.5.1.3.2 Control panel communications. Binary serial data shall be transmitted from the CPU TO CONTROL PANEL DATA port, J7-T (HI) and J7-U (LO), and binary serial data shall be received by the CONTROL PANEL TO CPU DATA port, J7-P (HI) and J7-N (LO) as differential RS-422 signals. Data for transmission shall be written to address 0A0078, and received data shall be read at address 0A0078.

3.5.1.3.2.1 Transmitting. The transmitted data provides either a four-digit message to an external control panel display, requests the input of external control panel switch position data, or initiates an end-around-test (EAT). Transmitted serial binary data shall be in the form of 8-bit data bytes, each with an additional start bit, two stop bits and an odd parity bit as shown in Figure 1. Each transmitted data byte shall be interpreted as shown in Table II. The transmitted data for writing to the four digits of the control panel display shall be converted as shown in Tables III and IV. Digit 1 of the display is the right-most and least significant digit (LSD), and Digit 4 is the left-most and most significant digit (MSD). The character "X" shown in Tables II and III denotes a logical "don't-care" condition; i.e., the data bit may be either a logic "0" or a logic "1".

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.



NOTE: The parity bit is odd; i.e., the sum of the 1's in each data byte plus its parity bit equals an odd number.

FIGURE 1. Data format for serial binary communications at 32.250 kHz with odd parity.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

TABLE II. Function definition of a Control Panel transmitted data byte.

FUNCTION	D15	D14	D13	D12	D11	D10	D9	D8
Write to digit 1 (LSD)	0	0	0	----See Table III----				
Write to digit 2	0	0	1	----See Table III----				
End-around test (EAT)	0	1	0	X	X	X	X	X
Write to digit 3	0	1	1	----See Table III----				
Write to digit 4 (MSD)	1	0	0	----See Table IV-----				
End-around test (EAT)	1	0	1	X	X	X	X	X
Read switch positions	1	1	0	X	X	X	X	X
No operation	1	1	1	X	X	X	X	X

TABLE III. Conversion codes for digits 1, 2 and 3.

D12	D11	D10	D9	D8	CHARACTER
1	0	0	0	0	0
X	0	0	0	1	1
X	0	0	1	0	2
X	0	0	1	1	3
X	0	1	0	0	4
X	0	1	0	1	5
X	0	1	1	0	6
X	0	1	1	1	7
X	1	0	0	0	8
X	1	0	0	1	9
0	0	0	0	0	Blank

Source: https://assist.dla:mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the current version before use.

TABLE IV. Conversion codes for digit 4.

D12	D11	D10	D9	D8	CHARACTER
1	0	0	0	0	0
1	0	0	0	1	1
1	0	0	1	0	2
1	0	0	1	1	3
1	0	1	0	0	4
1	0	1	0	1	5
1	0	1	1	0	6
1	0	1	1	1	7
1	1	0	0	0	8
1	1	0	0	1	9
0	0	0	0	0	Blank
1	1	0	1	0	A
1	1	0	1	1	C
1	1	1	0	0	E
1	1	1	0	1	F
1	1	1	1	0	-
1	1	1	1	1	U
0	0	0	0	1	=

NOTE: "=" denotes top and bottom segments.

3.5.1.3.2.2 Receiving. The received serial binary data shall be an 8-bit data byte plus a start bit, two stop bits and an odd parity bit as shown in Figure 1. The received data byte shall be encoded as shown in Table V, when switch position data is requested. The least significant four bits of the data byte encode the position of a multiposition FUNCTION switch (FSP) and the most significant four bits encode the position of four discrete switches:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

DISPLAY/ENTRY (DDE), SELECT (DSS), INCREMENT (DIS), and ENTER (ESP). These switch inputs provide a manual means to communicate with the FCP. The DISPLAY/ENTRY switch shall toggle the 4-digit display between display and entry modes. The SELECT switch shall be momentary and shall sequentially select each digit; the selected digit shall blink. The INCREMENT switch shall be momentary and shall sequentially increment the selected digit's contents. The ENTER switch shall be momentary and shall instruct the FCP to execute the centered 4-digit command. These switch inputs shall be activated in the sequence required by the FCP's self test BIT routines (Start up BIT (SUBI) and E910-E916).

TABLE V. Function definition of data byte received from Control Panel.

Switch Position	D15	D14	D13	D12	D11	D10	D9	D8
DISPLAY/ENTRY set to ENTRY	0							
DISPLAY/ENTRY set to DISPLAY	1							
ENTER depressed		0						
ENTER released		1						
INCR depressed			0					
INCR released			1					
SELECT depressed				0				
SELECT released				1				
FUNCTION set to OFF					0	0	0	0
FUNCTION set to LT					0	0	0	1
FUNCTION set to MV					0	0	1	0
FUNCTION set to AD					0	0	1	1
FUNCTION set to AT					0	1	0	0
FUNCTION set to WS					0	1	0	1
FUNCTION set to WD					0	1	1	0
FUNCTION set to T					0	1	1	1
FUNCTION set to CT					1	0	0	0
FUNCTION set to B					1	0	0	1
FUNCTION set to ZO					1	0	1	0
FUNCTION set to SP					1	0	1	1
FUNCTION set to Self Test					1	1	0	0

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

3.5.1.3.3 Distribution box, Unit 2 and Unit 5 communications. These three communication interfaces share the same UART, therefore, only one can be active at a time. Data for transmission shall be written to address 0A007C, and received data shall be read at address 0A007C.

3.5.1.3.3.1 Distribution box communications. Port CPU TO DISTRIBUTION BOX DATA, J8-S (HI) and J8-T (LO), shall transmit, and port DISTRIBUTION BOX TO CPU DATA, J8-U (HI) and J8-V (LO), shall receive binary serial data. The data shall comply with the EIA RS-422 standard and Figure 1.

3.5.1.3.3.2 Unit 2 communications. Port CPU TO UNIT 2 DATA, J7-J (HI) and J7-K (LO), shall transmit, and port UNIT 2 to CPU DATA, J7-L (HI) and J7-M (LO), shall receive binary serial data. The data shall comply with the EIA RS-422 standard and Figure 1.

3.5.1.3.3.3 Unit 5 communications. Port CPU TO UNIT 5 DATA, J6-D (HI) and J6-U (LO), shall transmit, and port UNIT 5 TO CPU DATA, J6-V (HI) and J6-W (LO), shall receive binary serial data. The data shall comply with the EIA RS-422 standard and Figure 1.

3.5.1.3.4 Unit 4 communications. Port CPU TO UNIT 4 DATA, J6-N (HI) and J6-M (LO), shall transmit, and port UNIT 4 TO CPU DATA, J6-L (HI) and J6-(D) (LO), shall receive binary serial data. The data shall comply with the EIA RS-422 standard and Figure 1, except two consecutive received data bytes shall load to form a 16-bit word (D0 through D15). Data for transmission shall be written to address 0A0074, and received data shall be read at address 0A0074.

3.5.1.3.5 Link #1 communications. Data Link #1 shall be capable of transmitting and receiving full 16-bit words per the EIA RS-422 standard and Figure 2. High and low bytes shall be transmitted simultaneously. Data for transmission shall be written to address 0C4000, and received data shall be read at address 0C4000. The data rate shall be 38.461 kHz ± 6 Hz, which equates to a UART clock frequency of 615.385 kHz. Ports shall be as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

High byte redundant output ports:
CPU TO DATA LINK #1 HIGH DATA, J3-A (HI) and J3-B (LO)
CPU TO DATA LINK #1 HIGH DATA, J3-C (HI) and J3-D (LO)

Low byte redundant output ports:
CPU TO DATA LINK #1 LO DATA, J3-(J) (HI) and J3-(K) (LO)
CPU TO DATA LINK #1 LO DATA, J3-G (HI) and J3-H (LO)

High byte input port:
DATA LINK #1 TO CPU HI DATA, J3-(F) (HI) and J3-(H) (LO)

Low byte input port:
DATA LINK#1 TO CPU LO DATA, J3-(I) (HI) and J3-(J) (LO)

Data Link #1 shall act as the DTOT dump and output the variables described in section 3.4.8 of the CPPS, when provided with the following inputs simultaneously:

a. Power per 3.5.1.1,

b. TTL logic low at $\overline{\text{RUN DTOT}}$, J3-AA,

c. +24 +6/-2 Vdc at:

(1) RADAR MODE, J1-(B), or

(2) MANUAL MODE, J1-E, or

(3) EXTERNAL MODE, J1-(J) and/or J1-(M), or

(4) GROUND MODE, J1-(P),

d. +24 +6/-2 Vdc at ACTION SWITCH, J1-A

A logic low applied to $\overline{\text{RUN TRAINING}}$, J3-BB, will be used to enable Link #1 to be bidirectional; i.e., to receive and send serial data when special software, not presently part of the FCP configuration, is added to the Computer and Memory Board EPROMs.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

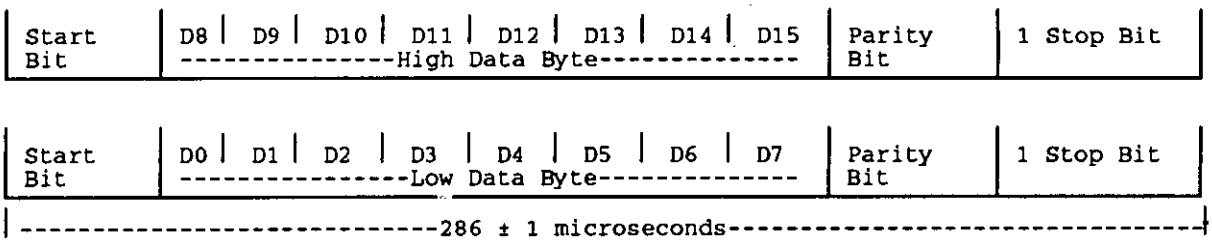


FIGURE 2. Link #1 data format for serial binary communications at 38.461 kHz with odd parity.

3.5.1.3.6 Link #2 and Link #3 communications. Links #2 and #3 share the same UART, therefore, only one of these links can be active at a time. Data for transmission shall be written to address 0C6000, and received data shall be read at address 0C6000.

3.5.1.3.6.1 Link #2. Data Link #2, known as the LEC link, is bidirectional and shall output data bus variables, when special software is added to the Data Transmission Board. This software is presently not part of the present FCP configuration. The Link #2 input and output selector circuits, output data latch, receiver, driver and UART in the present configuration shall be functional. Port CPU TO DATA LINK #2 DATA, J3-L (HI) and J3-M (LO), shall transmit, and port DATA LINK #2 TO CPU DATA, J3-N (HI) and J3-P (LO), shall receive binary serial data. The data shall comply with the EIA RS-422 standard and Figure 3. The data rate shall be 19.231 kHz ± 3 Hz, which equates to a UART clock frequency of 307.692 kHz. A TTL logic low applied to RUN LEC LINK, J3-EE, will enable Link #2, when special software is added.

3.5.1.3.6.2 Link #3. Data Link #3, known as the weather station link, is a bidirectional serial 8-bit circuit added for a future modification to input weather parameters from external sensors for more accurate ballistic computations and output these weather parameters to a display. The present configuration of the FCP does not include the data input/output and clock wiring from the Data Transmission Board to the FCP ports nor the special software to activate and utilize this circuitry as a "weather station". The Link #3 input and output selector circuits, output data latch and UART in the present configuration shall be functional. The data rate shall be 19.231 kHz ± 3 Hz, which equates to a UART clock frequency of 307.692 kHz. Serial data format shall be per Figure 3.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

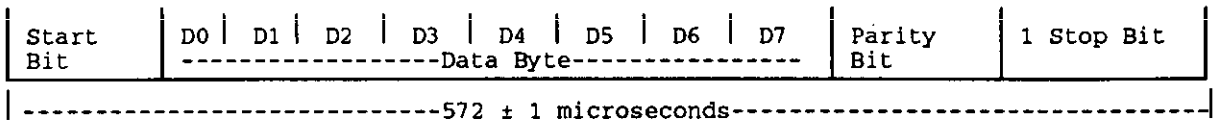


FIGURE 3. Link #2 and Link #3 data format for serial binary communications at 19.231 kHz with odd parity.

3.5.1.4 Continuity. Electrical continuity shall exist between the following connections:

a. ACTION SWITCH RELAY (J1-(K)) to FOOT SWITCH (J5-B) to ANTENNA ACTION SWITCH (J6-(G)).

b. EXTERNAL MODE (J1-(J)) to EXTERNAL MODE (J1-(M)).

c. RADAR OPERATE (J1-X) to RADAR OPERATE (J6-Y).

d. FIRE TRIGGER RELAY (J1-N) to FIRE TRIGGER RELAY (J6-X).

e. FOOT SWITCH (J5-A) to RADIATE COMMAND (J6-(H)).

3.5.1.5 High level discrete input signals. The following inputs shall be selectively set to a logic high "1" state by the application of +16 ± 1 Vdc or to a logic low "0" state by the application of 0 ± 1 Vdc. Pin J1-R shall be used as a common return for these inputs. The response, read on Data Bus 1/2 shall be as shown.

3.5.1.5.1 Logic high (+16 ± 1 Vdc).

3.5.1.5.1.1 GROUND MODE. With a 16 Vdc logic high applied to J1-(P),

a. bit D4 of the VMSP word shall be TTL logic high, when read at IOB address 0A0065.

b. bit D15 (GMB) of the SCMW word shall be TTL logic low, when read at SCB common memory address 404E.

3.5.1.5.1.2 RADAR MODE. With a 16 Vdc logic high applied to J1-(B),

a. bit D3 of the VMSP word shall be TTL logic high, when read at IOB address 0A0065.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the current version before use.

3.5.1.5.1.3 MANUAL MODE. With a 16 Vdc logic high applied to J1-E,

a. bit D2 of the VMSP word shall be TTL logic high, when read at IOB address 0A0065.

3.5.1.5.1.4 EXTERNAL MODE. With a 16 Vdc logic high applied to J1-(M) and/or J1-(J),

a. bit D1 of the VMSP word shall be TTL logic high, when read at IOB address 0A0065.

3.5.1.5.1.5 TEST MODE. With a 16 Vdc logic high applied to J1-(D),

a. bit D5 of the VMSP word shall be TTL logic high, when read at IOB address 0A0065.

3.5.1.5.1.6 ACTION SWITCH. With a 16 Vdc logic high applied to J1-A,

a. bit D0 (ACT) shall be TTL logic high, when read at IOB address 0A0041.

b. bit D14 of the SCMW word (ASB) shall be TTL logic low, when read at SCB common memory address 404E.

c. bit D5 (RSAS) shall be TTL logic low, when read at DTB address 0C7001.

3.5.1.5.1.7 ACQUISITION/TRACK SWITCH. With a 16 Vdc logic high applied to J1-(S),

a. bit D3 (AKTRK) shall be TTL logic high, when read at IOB address 0A0041.

b. bit D3 (RSATS) shall be TTL logic low, when read at DTB address 0C7001.

3.5.1.5.1.8 FAULT BYPASS. With a 16 Vdc logic high applied to J1-Z,

a. bit D7 (FBS) shall be TTL logic high, when read at IOB address 0A0065.

b. bit D13 of the SCMW word (FBB) shall be TTL logic low, when read at SCB common memory address 404E.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the currentversion before use.

3.5.1.5.1.9 FIRE TRIGGER RELAY. With a 16 Vdc logic high applied to J1-N,

a. bit D2 (GFTX) shall be TTL logic high, when read at IOB address 0A0041.

b. bit D2 (RSFTR) shall be TTL logic low, when read at DTB address 0C7001.

3.5.1.5.1.10 EXTERNAL RANGE ENABLE. With a 16 Vdc logic high applied to J1-(G),

a. bit D6 (ERE) shall be TTL logic high, when read at IOB address 0A0065.

3.5.1.5.1.11 FOOT SWITCH. With a 16 Vdc logic high applied to J5-A,

a. bit D1 (RADC) shall be TTL logic high, when read at IOB address 0A0041.

b. bit D4 (RSRC) shall be TTL logic low, when read at DTB address 0C7000.

3.5.1.5.2 Logic low (0 ± 1 Vdc).

3.5.1.5.2.1 GROUND MODE. With a 0 Vdc logic low applied to J1-(P),

a. bit D4 of the VMSP word shall be TTL logic low, when read at IOB address 0A0065.

b. bit D15 of the SCMW word (GMB) shall be TTL logic high, when read at SCB common memory address 404E.

3.5.1.5.2.2 RADAR MODE. With a 0 Vdc logic low applied to J1-(B),

a. bit D3 of the VMSP word shall be TTL logic low, when read at IOB address 0A0065.

3.5.1.5.2.3 MANUAL MODE. With a 0 Vdc logic low applied to J1-E,

a. bit D2 of the VMSP word shall be TTL logic low, when read at IOB address 0A0065.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

3.5.1.5.2.4 EXTERNAL MODE. With a 0 Vdc logic low applied to J1-(M) and/or J1-(J),

a. bit D1 of the VMSP word shall be TTL logic low, when read at IOB address 0A0065.

3.5.1.5.2.5 TEST MODE. With a 0 Vdc logic low applied to J1-(D),

a. bit D5 of the VMSP word shall be TTL logic low, when read at IOB address 0A0065.

3.5.1.5.2.6 ACTION SWITCH. With a 0 Vdc logic low applied to J1-A,

a. bit D0 (ACT) shall be TTL logic low, when read at IOB address 0A0041.

b. bit D14 of the SCMW word (ASB) shall be TTL logic high, when read at SCB common memory address 404E.

c. bit D5 (RSAS) shall be TTL logic high, when read at DTB address 0C7001.

3.5.1.5.2.7 ACQUISITION/TRACK SWITCH. With a 0 Vdc logic low applied to J1-(S),

a. bit D3 (AKTRK) shall be TTL logic low, when read at IOB address 0A0041.

b. bit D3 (RSATS) shall be TTL logic high, when at DTB address 0C7001.

3.5.1.5.2.8 FAULT BYPASS. With a 0 Vdc logic low applied to J1-Z,

a. bit D7 (FBS) shall be TTL logic low, when read at IOB address 0A0065.

b. bit D13 of the SCMW word (FBB) shall be TTL logic high, when read at SCB common memory address 404E.

3.5.1.5.2.9 FIRE TRIGGER RELAY. With a 0 Vdc logic low applied to J1-N,

a. bit D2 (GFT) shall be TTL logic low, when read at IOB address 0A0041.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

b. bit D2 (RSFTR) shall be TTL logic high, when read at DTB address 0C7001.

3.5.1.5.2.10 EXTERNAL RANGE ENABLE. With a 0 Vdc logic low applied to J1-(G),

a. bit D6 (ERE) shall be TTL logic low, when read at IOB address 0A0065.

3.5.1.5.2.11 FOOT SWITCH. With a 0 Vdc logic low applied to J5-A,

a. bit D1 (RADC) shall be TTL logic low, when read at IOB address 0A0041.

b. bit D4 (RSRC) shall be TTL logic high, when read at DTB address 0C7000.

3.5.1.6 Low level (TTL) discrete input signals. The following inputs are standard Transistor-Transistor Logic (TTL) compatible, and they shall be selectively set to a logic high "1" state by application of 2.4 to 5.2 Vdc or to a logic low "0" state by application of 0.0 to 0.4 Vdc. Pin J3-GG shall be used as a common return for these inputs. The response, read on Data Bus 1/2, shall be as shown.

3.5.1.6.1 Logic high (+5 V).

3.5.1.6.1.1 $\overline{\text{RUN DTOT}}$. With a TTL logic high applied to J3-AA,

a. bit D9 (DTLNKON) shall be TTL logic high, when read at DTB address 0C3000.

3.5.1.6.1.2 $\overline{\text{RUN TRAINING}}$. With a TTL logic high applied to J3-BB,

a. bit D8 (DTTRNON) shall be TTL logic high, when read at DTB address 0C3000.

3.5.1.6.1.3 $\overline{\text{RUN LEC LINK}}$. With a TTL logic high applied to J3-EE,

a. bit D1 (WSLO) shall be TTL logic high, when read at DTB address 0C5000.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the current version before use.

3.5.1.6.1.4 RUN FCP UNIT TEST. With a TTL logic high applied to J3-CC,

a. bit D1 (DTFUT) shall be TTL logic high, when read at DTB address 0C3000.

3.5.1.6.1.5 SP/TWD. With a TTL logic high applied to J8-GG,

a. bit D12 of the SCMW word (TOWSP) shall be TTL logic high, when read at SCB common memory address 404E.

3.5.1.6.2 Logic low (OV).

3.5.1.6.2.1 RUN DTOT. With a TTL logic low applied to J3-AA,

a. bit D9 (DTLNKON) shall be TTL logic low, when read at DTB address 0C3000.

3.5.1.6.2.2 RUN TRAINING. With a TTL logic low applied to J3-BB,

a. bit D8 (DTTRNON) shall be TTL logic low, when read at DTB address 0C3000.

3.5.1.6.2.3 RUN LEC LINK. With a TTL logic low applied to J3-EE,

a. bit D1 (WSLO) shall be TTL logic low, when read at DTB address 0C5000.

3.5.1.6.2.4 RUN FCP UNIT TEST. With a TTL logic low applied to J3-CC,

a. bit D1 (DTFUT) shall be TTL logic low, when read at DTB address 0C3000.

3.5.1.6.2.5 SP/TWD. With a TTL logic low applied to J8-GG,

a. bit D12 of the SCMW word (TOWSP) shall be TTL logic low, when read at SCB common memory address 404E.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

3.5.1.7 Discrete output signals.

3.5.1.7 COMPUTER POWER LAMP. J1-K shall go logic low to +24 V (BAT) RTN, when power is applied to the FCP per 3.5.1.1. The COMPUTER POWER LAMP output (J1-K) shall be capable of sinking 50 milliamperes maximum at 24 ± 2 Vdc. When the voltage of one or more of the following FCP input power supplies falls below the following limits, the COMPUTER POWER LAMP output at J1-K shall go TTL logic high.

Supply	Limits
+5 V	+4.5 ± 0.3 V
+15 V	+13.0 ± 0.9 V
-15 V	-13.0 ± 0.4 V

3.5.1.7.2 COMPUTER CYCLE. J1-V shall go logic low to +24 V (BAT) RTN, when power is applied to the FCP per 3.5.1.1, to show the Computer and Memory Board is receiving the 64 Hz interrupt from the Servo Controller Board. The COMPUTER CYCLE output (J1-V) shall be capable of sinking 50 milliamperes maximum at 24 ± 2 Vdc.

3.5.1.7.3 ACTION SWITCH relay. With a logic high ACTION SWITCH input applied to J1-A per 3.5.1.5.1.6, the ACTION SWITCH relay's normally open contacts between J3-Y and J3-Z shall close and be capable of conducting 1.0 ampere maximum (resistive) at 28 Vdc.

3.5.1.7.4 ACTION SWITCH LOCKOUT. With a TTL logic high written to bit D0 (SALO) at address 0A0053 (SALO1) with no fault bypass switch input:

a. the ACTION SWITCH LOCKOUT output at J1-(A) shall transition logic low to the +24 V (BAT) RTN at J1-R and be capable of sinking 40 milliamperes maximum.

b. bit D4 (SALO1) at address 0A0041 (SWTCHIN2) shall transition TTL logic high.

3.5.1.7.5 READY TO FIRE LAMP and relay. With a TTL logic high written to bit D0 at address 0A0059 (RTFLAMP):

a. the READY TO FIRE LAMP output at J4-P shall transition logic low to the +24 V (BAT) RTN at J1-R and be capable of sinking 50 milliamperes maximum.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

b. the normally open relay contacts between J3-(A) and J3-(B) shall transition from open to closed. The relay contacts shall be capable of conducting a current of 1 ampere (resistive) at 28 Vdc.

c. bit D7 (RSRTF) shall be TTL logic low when read at DTB address 0C7000.

3.5.1.7.6 FAULT LAMP. With a TTL logic high written to bit D0 at address 0A0055 (SFLAMP), the FAULT LAMP output at J1-(E) shall transition logic low to the +24 V (BAT) RTN at J1-R and be capable of sinking 50 milliamperes maximum.

3.5.1.7.7 $\overline{\text{TRACK/JAM LAMP}}$ and relay. With a TTL logic high written to bit D0 at address 0A0057 (TRKJAM):

a. the $\overline{\text{TRACK/JAM LAMP}}$ output at J4-S shall transition logic low to the +24 V (BAT) RTN at J1-R and be capable of sinking 50 milliamperes maximum.

b. the TRACK/JAM relay output at J3-F shall transition logic high to output +24 V (BAT) with respect to the +24 V (BAT) RTN at J1-R and be capable of sourcing 1 ampere maximum (resistive) at 28 Vdc.

c. bit D6 (RSTJ) shall be TTL logic low when read at DTB address 0C7000.

3.5.1.7.8 ACQ/TRACK SWITCH relay. With +16 ± 1 Vdc applied to J1-(S) with respect to J1-R as return, the normally closed relay contacts between J3-S and J3-R shall open and the normally open relay contacts between J3-S and J3-T shall close. These relay contacts shall be capable of conducting a current of 1 ampere maximum (resistive) at 28 Vdc.

3.5.1.7.9 RADIATE COMMAND relay. With +16 ± 1 Vdc applied to J5-A with respect to J1-R as return, the normally open relay contacts between J3-U and J3-V shall close. These relay contacts shall be capable of conducting a current of 1 ampere maximum (resistive) at 28 Vdc.

3.5.1.7.10 FIRE TRIGGER RELAY relay. With +16 ± 1 Vdc applied to J1-N with respect to J1-R as return, the normally closed relay contacts between J3-(C) and J3-(G) shall open and the normally open relay contacts between J3-(C) and J3-(E) shall close. These relay contacts shall be capable of conducting a current of 1 ampere maximum (resistive) at 28 Vdc.

Source: https://assist.dla.mil -- Downloaded: 20260-26T04:29Z
Check the source to verify that this is the current version before use.

3.5.1.8 Analog input signals. The following analog input signals shall be converted to digital and read from random access memory (RAM), after processing by the FCP's Servo Controller Board (SCB) or Input/Output Board (IOB).

3.5.1.8.1 SCB scaled, selected and converted input signals. Each analog input signal of Table VI, applied to the FCP input connector-pins shown, shall be hardware scaled, then automatically sequentially multiplexer selected, converted to a complementary-two's-complement digital word by a 12-bit analog to digital converter (ADC), and loaded into a first-in first-out (FIFO) shift register. SCB module IADINPUT shall write to address 8040 sixteen times to sequentially shift out each 12-bit word through inverting buffers (forming two's-complement logic) onto the data bus and into respective SCP RAM locations per Table VI. (Four logic high LSBs are added to the 12-bit ADC output, prior to buffer inversion, to form 16-bit words with the 4 LSBs logic low). Each stored 16-bit word shall represent (within the channel maximum error tolerance shown) its respective applied input voltage scaled by the nominal hardware scale factor given in the table, and a 16-bit ADC conversion factor of 0.3052 mV/LSB. (The conversion factor for the 12-bit ADC is 4.882813 mV/LSB.) The TACH signals have additional software scaling factors per Note 3 at the end of the table. Each digital word shall be updated every 15.6 milliseconds (at a 64 Hertz rate).

TABLE VI. SCB analog input signals.

Signal Name; Variable Name; Type Signal	Signal Voltage (in volts)	FCP Input Connector-Pin	Nominal Scale Factor & SCB RAM Address	Channel No. & Max. Error
GROUND (Grounded within FCP)	0 ± 0.2	None	x1 [1] None	0 N.A.
HAND CONTROL AZ; AZSTKX1; DC, Differential	-15 to +15	J1-P (HI) J1-F (LO)	x(-1.083) 4118	1 ± 0.8%
HAND CONTROL AZ; AZSTKX8; DC, Differential	-15 to +15	J1-P (HI) J1-F (LO)	x(-8.667) [2] 411A	2 ± 0.8%
GUN TACH AZ; AZTACHX1; DC, Differential	-15 to +15	J8-A (HI) J8-B (LO)	x(-0.326) [3] 411C	3 ± 0.8%
GUN TACH AZ; AZTACHX8; DC, Differential	-15 to +15	J8-A (HI) J8-B (LO)	x(-2.608) [3] 411E	4 ± 1.0%
HAND CONTROL EL; ELSTKX1; DC, Differential	-15 to +15	J1-D (HI) J1-(N) (LO)	x(-1.083) 4120	5 ± 0.8%

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the current version before use.

TABLE VI. SCB analog input signals. (Continued)

Signal Name; Variable Name; Type Signal	Signal Voltage (in volts)	FCP Input Connector-Pin	Nominal Scale Factor & SCB RAM Address	Channel No. & Max. Error
HAND CONTROL EL; ELSTKX8; DC, Differential	-15 to +15	J1-D (HI) J1-(N) (LO)	x(-8.667) [2] 4122	6 ± 0.8%
GUN TACH EL; ELTACHX1; DC, Differential	-15 to +15	J8-(X) (HI) J8-(Y) (LO)	x(-0.155) [3] 4124	7 ± 0.8%
GUN TACH EL; ELTACHX8; DC, Differential	-15 to +15	J8-(X) (HI) J8-(Y) (LO)	x(-1.240) [3] 4126	8 ± 1.0%
SPARE (grounded within FCP)	0 ± 0.2	None	x1 4128	9 ± 0.6%
SPARE (grounded within FCP)	0 ± 0.2	None	x1 412A	10 ± 0.6%
MANUAL/EXTERNAL [4] RANGE; RANGE; DC	0 to +15	J1-(H) J1-U (RTN)	x0.667 412C	11 ± 2.0%
CURRENT AZ 1; [5] AI1; DC	-15 to +15	J8-D (HI) J8-E (LO)	x1 412E	12 ± 0.6%
CURRENT AZ 2; [5] AI2; DC	-15 to +15	J8-J (HI) J8-k (LO)	x1 4130	13 ± 0.6%
CURRENT EL; [5] EI; DC	-15 to +15	J8-(D) (HI) J8-(E) (LO)	x1 4132	14 ± 0.6%

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the current version before use.

TABLE VI. SCB analog input signals. (Continued)

Signal Name; Variable Name; Type Signal	Signal Voltage (in volts)	FCP Input Connector-Pin	Nominal Scale Factor & SCB RAM Address	Channel No. & Max. Error
COMPOSITE SCP RATE COMMANDS; D/AEAT; DC This signal is the sum of FCP outputs:	-10 to +10	None	x(-1) 4134	15 [6] ± 3.2%
DAC 4: SIGHT CMD AZ; VD/AAZS; DC		None (Internally connected to DAC output)		
DAC 3: SIGHT CMD EL; VD/AELS; DC		None (Internally connected to DAC output)		
DAC 2: GUN CMD AZ; VD/AAZG; DC		None (Internally connected to DAC output)		
DAC 1: GUN CMD EL; VD/AELG; DC		None (Internally connected to DAC output)		

NOTES:

1. Ground is not output to RAM, but instead represents ADC offset error, which is subtracted by FCP software from all other input conversions.
2. AZSTKX8 and ELSTKX8 are clipped at 8.6 Vdc by software.
3. In addition to the hardware scale factor shown for each tachometer input, there are software scale factors as follows:
   AZTACHX1 and AZTACX8 are both software scaled by -1.125 and:
   for the self-propelled system by 1.0.
   for the towed system by 0.9089.
   ELTACHX1 and ELTACHX8 are both software scaled by -1.11 and 0.90524.
4. MANUAL/EXTERNAL RANGE may be converted from a voltage (V) in volts to a range in meters using: RANGE = (317.45 x V) - 20.
5. The current inputs may be converted from a voltage to a current, using the relationship: 15 amperes/volt.
6. The error tolerance shown for channel 15 is from the digital inputs of the 4 DACs to the SCB ADC digital output stored in RAM. Not all SCB DAC analog outputs (SCB EAT analog inputs) are available externally at FCP connectors, therefore, the channel 15 analog inputs are generated by writing digital words to the SCB DACs.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the current version before use.

3.5.1.8.2 IOB scaled, selected and converted input signals. The analog input signals of Table VII shall be hardware scaled, multiplexer selected, converted to digital by an ADC, software scaled, and read from the low data byte at IOB address 0A0061 (ADDATA). Input signal type, voltage and hardware scale factor shall be per Table VII. The software scale factor shall be "x5". These signals shall be converted by a 10-bit ADC, but only the 8 most significant bits (MSBs) shall be applied to the low byte of the 16-bit data bus. (The 2 LSBs of the ADC are not used.) The ADC's conversion factor is 39.0625 mV/LSB for the used bits. The resulting data on the bus shall be positive-true offset binary logic per paragraph 6.7. IOB address 0A0061 shall be read (and the read data ignored) to reset the ADC. Each input channel shall be selected by writing its corresponding channel number (per Table VII) to the low byte of IOB address 0A006D (CHANSLT). The analog to digital conversion shall be initiated by writing any low byte data to IOB address 0A0040 (STADCONV). The conversion shall be complete when bit D7 (A DATA READY) of the status byte at IOB address 0A0041 (SWTCHIN2) goes logic high.

TABLE VII. IOB analog input signals.

Signal Name; Variable Name; Type Signal	Signal Voltage (in volts)	PCP Input Connector-Pin	Nominal HW Scale Factor	Channel No. & Max. Error [1]
GUN COMMAND EL; None; DC	-10 to +10	None (Internally connected to Gun El output)	x2.004	00 ± 5.1%
GUN COMMAND AZ 1 & 2; None; DC	-10 to +10	None (Internally connected to Gun Az output)	x1.000	01 ± 5.1%
115 Vrms 400 Hz REF; None; AC	115 ± 5%	J2-M (R1) J2-K (R2)	x0.0354	02 ± 4.4%
RANGE VOLTAGE; [2] None; DC	0 to +10	None (Internally connected to a DAC output)	x0.440	03 ± 3.5%
+24 V; None; DC	+24.0 ± 1.2	J2-H J2-R (RTN)	x0.142	04 ± 4.2%
-24 V; None; DC	-24.0 ± 1.2	J2-J J2-R (RTN)	x0.142	05 ± 4.2%

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the current version before use.

TABLE VII. IOB analog input signals. Continued

Signal Name; Variable Name; Type Signal	Signal Voltage (in volts)	FCP Input Connector-Pin	Nominal HW Scale Factor	Channel No. & Max. Error [1]
MANUAL SPEED; [3] TSPP; DC	0 to +15	J1-T J1-U (RTN)	x0.440	06 ± 3.5%
-15 V; None; DC	-15.00 ± 0.75	J2-B J2-N (RTN)	x0.282	07 ± 3.9%
+15 V; None; DC	+15.00 ± 0.75	J2-A J2-N (RTN)	x0.282	08 ± 3.9%
AZ MOT 1 CW; None; DC	0 to +24	J8-AA J8-HH (RTN)	x0.142	09 ± 4.2%
AZ MOT 1 CCW; None; DC	0 to +24	J8-BB J8-HH (RTN)	x0.142	0A ± 4.2%
AZ MOT 2 CW; None; DC	0 to +24	J8-CC J8-HH (RTN)	x0.142	0B ± 4.2%
AZ MOT 2 CCW; None; DC	0 to +24	J8-DD J8-HH (RTN)	x0.142	0C ± 4.2%
EL MOT UP; None; DC	0 to +24	J8-EE J8-HH (RTN)	x0.142	0D ± 4.2%
EL MOT DOWN; None; DC	0 to +24	J8-FF J8-HH (RTN)	x0.142	0E ± 4.2%
+5 V; None; DC	+5.1 ± 0.255	J2-D,E J2-P,F (RTN)	x0.833	0F ± 2.7%

NOTES:

1. Allowable error with zero input for all channels is ± 20 mV, due to the bipolar offset characteristic of the IOB's ADC.
2. RANGE VOLTAGE may be converted from a voltage (v) in volts to a range in meters using: RANGE = 512 x V.
3. MANUAL SPEED may be converted from a voltage (V) in volts to a speed in knots using: MANUAL SPEED = (-194 x V) + 625.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the current version before use.

3.5.1.9 Analog output signals.

3.5.1.9.1 GUN COMMAND EL output. Analog voltages, representing commanded gun elevation rate, shall be output at J8-(F) (HI) and J8-(G) (LO) via a digital to analog converter (DAC). The digital word input to the Gun Command El DAC, VD/AELG, shall be in offset binary, but shall be obtained as software variable SGEC in two's complement binary from servo control processor (SCP) main program RAM address 4104. Software module IDAOTPUT shall convert this 16-bit word from two's complement to offset binary and write it to the Gun Command El DAC at address 801C. SGEC shall be computed by the reiterative algorithms given in the CPPS, except in BIT mode 2, where it is set equal to BIT variable GREC, located in common memory RAM at address 401A. The IHALTMOD software module shall write GREC from common memory address 401A to SCP main program memory address 4104, during the SCP 64 Hz interrupt. GUN COMMAND EL output voltage shall be limited to between -1.5 and +1.5 Vdc by software module GRATELES in all FCP modes except BIT Mode 2, where output voltages range between -10 and +10 Vdc. DAC conversion error shall be ±64.6 mV maximum. Zero offset error shall be ±20 mV maximum. One DAC LSB shall equal 0.3052 mV and a commanded elevation rate of 0.03723 milliradian/second.

3.5.1.9.2 GUN COMMAND AZ 1 AND GUN COMMAND AZ 2 outputs. Analog voltages, representing commanded gun azimuth rate, shall be output at J8-H (HI), J8-G (LO) and J8-N (HI), J8-M (LO) via a DAC through an operational amplifier low-pass filter. (The two azimuth outputs shall be internally hardwired together.) The digital word input to the Gun Command Az DAC, VD/AAZG, shall be in offset binary, but shall be obtained as software variable STAC in two's complement binary from SCP main program RAM address 4106. Software module IDAOTPUT shall convert this 16-bit word from two's complement to offset binary and write it to the Fun Command Az DAC at address 801E. STAC shall be computed by the reiterative algorithms given in the CPPS, except in BIT mode 2, where it is set equal to BIT variable GRAC, located in common memory RAM at address 401C. The IHALTMOD software module shall write GRAC from common memory address 401C to SCP main program memory address 4106, during the SCP 64 Hz interrupt. GUN COMMAND AZ 1 and GUN COMMAND AZ 2 output voltages shall be limited to between -3.0 and +3.0 Vdc by software module GRATELES in all FCP modes except BIT Mode 2, where output voltages range between -10 and +10 Vdc. DAC conversion error plus op-amp offset error shall be ±69.6 mV maximum. Zero offset error shall be ±25 mV maximum. One DAC LSB shall equal 0.3052 mV and a commanded azimuth rate of 0.03784 milliradian/second.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

3.5.1.9.3 SIGHT COMMAND EL output. Analog voltages, representing commanded sight elevation rate, shall be output at J4-(E) (HI), J4-M (LO) via a DAC. The digital word input to the Sight Command El DAC, VD/AELS (called SRECO in the CPPS), shall be in offset binary, but shall be obtained from SCP main program RAM address 4100 in two's complement binary. Software module IDAOTPUT shall convert this 16-bit word from two's complement to offset binary and write it to the Sight Command El DAC at address 8018. SRECO shall be computed by the reiterative algorithms given in the CPPS, except in BIT Mode 2, where it is set equal to BIT variable SREC, located in common memory RAM at address 4016. The IHALTMOD software module shall write SREC from common memory address 4016 to SCP main program memory address 4100, during the SCP 64 Hz interrupt. SIGHT COMMAND EL output voltages shall be between -10 and +10 Vdc. DAC conversion error shall be ±54.6 mV maximum. Zero offset error shall be ±20 mV maximum. One DAC LSB shall equal 0.3052 mV and a commanded rate of 0.03052 milliradian per second.

3.5.1.9.4 SIGHT COMMAND AZ output. Analog voltages, representing commanded sight azimuth rate, shall be output at J4-(F) (HI), J4-N (LO) via a DAC. The digital word input to the Sight Command Az DAC, VD/AAZS (called SRACO in the CPPS), shall be in offset binary, but shall be obtained from SCP main program RAM address 4102 in two's complement binary. Software module IDAOTPUT shall convert this 16-bit word from two's complement to offset binary and write it to the Sight Command Az DAC at address 801A. SRACO shall be computed by the reiterative algorithms given in the CPPS, except in BIT Mode 2, where it is set equal to BIT variable SRAC, located in common memory RAM at address 4018. The IHALTMOD software module shall write SRAC from common memory address 4018 to SCP main program memory address 4102, during the SCP 64 Hz interrupt. SIGHT COMMAND AZ output voltages shall be between -10 to +10 Vdc. DAC conversion error shall be ±54.6 mV maximum. Zero offset error shall be ±20 mV maximum. One DAC LSB shall equal 0.3052 mV and a commanded rate of 0.03052 milliradian per second.

3.5.1.9.5 RANGE VOLTAGE output. Digital 16-bit "Range and Mode" words shall be serially received at port UNIT 4 TO CPU DATA, J6-L (HI), J6-(D) LO and converted to parallel. Latched bits D11 through D4 (the 8 most-significant bits of the range part of each received word) shall be converted to an analog voltage by a DAC and output at J4-(G) HI and J4-(H) LO by writing XXXX000X (D7 through D0) to address 0A0071. The output range voltage in volts shall be given by the following expression:

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the current version before use.

RANGE VOLTAGE = [8-bit (D11 through D4) binary range converted to decimal number of LSBs x 0.0390625 V/LSB] ± 0.2 Vdc.

DAC conversion time shall be 3.0 microseconds maximum. Analog output voltage may be converted to range in meters using the following expression:

RANGE = (V_{Output}/ 0.0390625 V/LSB x 20 meters/LSB) ± 106.25 meters

3.5.1.10 Gun command overrides. The gun commands from the DAC outputs shall be interrupted by activating gun limit relays. The relay contacts are wired to open the gun command lines from the DACs and connect tachometer signals, instead, to the FCP's gun command output connectors. In normal system operation the tachometer output signals shall have a polarity opposite that of the interrupted DAC outputs to provide dynamic braking of the gun. Tachometer voltages in the range of -15 to +15 Vdc shall be input to the FCP at the following connector-pins:

GUN TACH EL: J8-(X) (HI), J8-(Y) (LO)
GUN TACH AZ: J8-A (HI), J8-B (LO)

3.5.1.10.1 DOWN LIMIT. The DOWN LIMIT relay shall be activated by the application of 23 ± 7 Vdc to J8-Y (+) and J8-Z (RTN). With the DOWN LIMIT relay activated, the GUN COMMAND EL output shall be given by:

$V_{GUN\ CMD\ EL}$ = -0.35(|$V_{EL\ TACH}$| - 0.47 V) ± 10 percent for negative tachometer voltages. $V_{EL\ TACH}$ shall be limited to 8.6 Vdc maximum by software.

$V_{GUN\ CMD\ EL}$ = 0 ± 0.1 Vdc for positive tachometer voltages

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

3.5.1.10.2 UP LIMIT. The UP LIMIT relay shall be activated by the application of 23 ± 7 Vdc to J8-X (+) and J8-Z (RTN). With the UP LIMIT relay activated, the GUN COMMAND EL output shall be given by:

$V_{GUN\ CMD\ EL}$ = +0.35($V_{EL\ TACH}$ - 0.47 V) ± 10 percent for positive tachometer voltages. $V_{EL\ TACH}$ shall be limited to 8.6 Vdc maximum by software.

$V_{GUN\ CMD\ EL}$ = 0 ± 0.1 Vdc for negative tachometer voltages

3.5.1.10.3 CCW LIMIT. The CCW LIMIT relay shall be activated by the application of 23 ± 7 Vdc to J8-(B) (+) and J8-C (RTN). With the CCW LIMIT relay activated, the GUN COMMAND AZ 1 and GUN COMMAND AZ 2 outputs shall be given by:

$V_{GUN\ CMD\ AZ}$ = -0.294(|$V_{AZ\ TACH}$| - 0.48) ±10 percent for negative tachometer voltages. $V_{AZ\ TACH}$ shall be limited to 8.6 Vdc maximum by software.

= 0 ± 0.1 Vdc for negative tachometer voltages

3.5.1.10.4 CW LIMIT. The CW LIMIT relay shall be activated by the application of 23 ± 7 Vdc to J8-A (+) and J8-(C) (RTN). With the CW LIMIT relay activated, the GUN COMMAND AZ 1 and GUN COMMAND AZ 2 outputs shall be given by:

$V_{GUN\ CMD\ AZ}$ = 0.294($V_{AZ\ TACH}$ - 0.48) ± 10 percent for positive tachometer voltages. $V_{AZ\ TACH}$ shall be limited to 8.6 Vdc maximum by software.

= 0 ± 0.1 Vdc for negative tachometer voltages

3.5.1.11 Sight and gun synchro input signals. The FCP shall convert the three-phase synchro input signals of Table VIII to digital angles within the tolerances given in the table. The converted angles shall be read at the SCP addresses shown. Each converted signal shall be a 16-bit word on data lines D15 through D0 with bit weighting as shown in Table IX. The converted data shall be latched within 3 microseconds of receiving the read instruction. For 1 LSB accuracy, a 179-degree step change shall require 264 milliseconds maximum settling time and a change less than or equal to 1.4 degrees shall require 35 milliseconds maximum settling time before being read.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

TABLE VIII. Synchro inputs.

Synchro Input Signal (0 to 360 degree phase angle)	Synchro Input Connector-Pin	Synchro to Digital Angle Conversion Tolerance	SCP Address for Reading (hexadecimal)
SIGHT SYNCHRO EL (3 Phase, 90 ± 27 Vrms L-L, 400 ± 0.4 Hertz)	S1: J4-(B) S2: J4-(C) S3: J4-H	± 0.02167 degrees (± 1.3 arc-minutes)	8032
SIGHT SYNCHRO AZ (3 Phase, 90 ± 27 Vrms L-L, 400 ± 0.4 Hertz)	S1: J4-F S2: J4-G S3: J4-H	± 0.02167 degrees (±1.3 arc-minutes)	8034
GUN SYNCHRO EL (3 Phase, 90 ± 27 Vrms L-L, 400 ± 0.4 Hertz)	S1: J7-D S2: J7-E S3: J7-(A)	± 0.08667 degrees (± 5.2 arc-minutes)	8036
GUN SYNCHRO AZ (3 Phase, 90 ± 27 Vrms L-L, 400 ± 0.4 Hertz)	S1: J7-A S2: J7-B S3: J7-C	± 0.08667 degrees (± 5.2 arc-minutes	8038

TABLE IX. Digital angle bit weighting.

Data Bus Bit (active logic high)	Weight (in angular degrees)	Weight (in radians or milliradians)
MSB D15	180	3.1415927 rad
D14	90	1.5707933
D13	45	0.7853982
D12	22.5	0.3926991
D11	11.25	0.1963495
D10	5.625	0.0981748
D9	2.8125	0.0490874
D8	1.40625	0.0245437
D7	0.703125	12.27184 mrad
D6	0.3515625	6.13592
D5	0.1757813	3.06796
D4	0.0878906	1.53398
D3	0.0439453	0.76699
D2	0.0219727	0.38350
D1	0.0109863	0.19175
LSB D0	0.0054932	0.09587

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the currentversion before use.

3.5.1.12 Antenna synchro output commands. The FCP shall convert 16-bit binary antenna position angle commands to three-phase synchro voltages and output them at the connector-pins shown in Table X. The digital to synchro conversion shall be accomplished by a digital-to-resolver converter (DRC) followed by a Scott-T transformer, which converts the resolver format into synchro format. The digital angle to be output shall be read from the SCP RAM address shown and written to the DRC address also shown in Table X. Binary weighting shall be per Table IX. Digital to synchro conversion time shall be 40 microseconds maximum to a conversion accuracy of ± 0.16667 degrees (10.0 arc-minutes).

TABLE X. Antenna synchro output commands.

Synchro Output Signal (0 to 360 degree phase angle)	Synchro Output Connector-Pin	SCP RAM Address (in hex)	DRC Address (in hex)
ANTENNA SYNCHRO EL (3 Phase, 90 ± 9 Vrms L-L, 400 ± 0.4 Hertz)	S1: J4-A S2: J4-B S3: J4-C	410A	8022
ANTENNA SYNCHRO AZ (3 Phase, 90 ± 9 Vrms L-L, 400 ± 0.4 Hertz	S1: J4-X S2: J4-Y S3: J4-Z	4108	8020

3.5.2 Spare connector-pins. The following connector-pins are designated as spares and shall not be connected:

J3-(D)
J3-(K)
J3-(M)
J3-DD
J6-(J)

3.6 First article. When specified in the contract or purchase order (see 6.2), a sample shall be subjected to first article inspection in accordance with the technical provisions herein (see 4.3).

3.7 Soldering. All soldering shall be in accordance with the requirements of Note 2 of Drawing 12950877.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the contractor is responsible for the performance of all inspection requirements (examinations and tests) as specified herein. Except as otherwise specified in the contract or purchase order, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in this specification where such inspections are deemed necessary to ensure supplies and services conform to prescribed requirements.

4.1.1 Responsibility for compliance. All items shall meet all requirements of Sections 3 and 5. The inspection set forth in this specification shall become a part of the contractor's overall inspection system or quality program. The absence of any inspection requirements in the specification shall not relieve the contractor of the responsibility of ensuring that all products or supplies submitted to the Government for acceptance comply with all requirements of the contract. Sampling inspection, as part of manufacturing operations, is an acceptable practice to ascertain conformance to requirements, however, this does not authorize submission of known defective material, either indicated or actual, nor does it commit the Government to accept defective material.

4.1.2 General provisions. Unless otherwise specified herein, the provisions of MIL-F-13926 apply and form a part of this specification. Reference shall be made to MIL-STD-109 to define quality assurance terms used herein.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3).

b. Quality conformance inspection (see 4.4).

c. Special sampling inspection (see 4.5).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

4.2.1 Inspection conditions. Unless otherwise specified, all examinations and tests shall be conducted at the following standard ambient conditions:

a.	Temperature	+50°F to +90°F
b.	Relative humidity	90 percent RH Max.
c.	Pressure	28 to 32 inches Hg

4.3 First article inspection.

4.3.1 Submission. The contractor shall submit a first article sample as designated by the Contracting Officer for evaluation in accordance with provisions of 4.3.2. The first article sample shall consist of the assemblies, components and test specimens listed below in the quantities indicated.

Name	Drawing	Quantity
FCP Assembly	9360890	3

4.3.2 Inspections to be performed. As determined by the Government, the first article assemblies, components and test specimens may be subjected to any or all of the examinations and tests specified in this detail specification and be inspected for compliance with any or all requirements of the applicable drawings.

4.3.3 Rejection. If any assembly, component or test specimen fails to comply with any of the applicable requirements, the first article sample shall be rejected. The Government reserves the right to terminate inspection upon any failure of an assembly, component or test specimen to comply with any of the requirements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

4.4 Quality conformance inspection.

4.4.1 Inspection lot formation. The term "inspection lot" is defined as a homogeneous collection of units of product from which a representative sample is drawn or which is inspected 100 percent to determine conformance with applicable requirements. Units of product selected for inspection shall represent only the inspection lot from which they are drawn and shall not be construed to represent any prior or subsequent quantities presented for inspection. Homogeneity shall be considered to exist provided the inspection lot has been produced by one manufacturer, in one unchanged process, using the same materials and methods, in accordance with the same drawings, same drawing revisions, same specifications and same specification revisions. All material submitted for inspection in accordance with this specification shall comply with the homogeneity criteria specified herein, regardless of the type of inspection procedure which is being applied to determine conformance with requirements.

4.4.2 Examinations and tests.

4.4.2.1 Components, assemblies and subassemblies.

a. The contractor's quality program or detailed inspection plan shall provide assurance of compliance of all characteristics with the applicable drawings and specifications utilizing as a minimum the conformance criteria specified in this document and the Quality Assurance Provisions (QAP) listed in the Technical Data Package (TDP). In the absence of QAPs, the applicable quality assurance provisions of MIL-F-13926 shall apply.

b. Unless otherwise specified herein or provided for in the contract, alternative quality conformance procedures, methods or equipment, such as statistical process control, tool control, other types of sampling plans, etc., may be used by the contractor when they provide, as a minimum, the level of quality assurance required by the provisions herein. Prior to applying such alternative procedures, methods or equipment, the contractor shall describe them in a written proposal submitted to the Government for evaluation (see 6.3). When required, the contractor shall demonstrate that the effectiveness of each proposed alternative is equal to or better than the specified quality conformance provisions(s) herein. In case of dispute as

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

to whether the contractor's proposed alternative(s) provides equivalent assurance, the provisions of this specification shall apply. All approved alternative provisions shall be specifically incorporated into the contractor's quality program or inspection system, as applicable.

4.4.2.2 Final acceptance inspection. Subsequent to first article approval, examinations and tests related to Section 3 shall be performed as specified in 4.4.2.3 and 4.5.

4.4.2.3 Functional tests. The tests in Table XI shall be performed on a 100 percent basis.

TABLE XI. Functional tests (100 percent inspection).

Characteristic	Requirement	Test Procedures
Critical	None	N/A
Major 101. Electrical 102. Performance	 3.4 3.5	 4.7.2 4.7.3
Minor	None	N/A

4.5 Special sampling. One Fire Control Processor out of the first 50 produced and one out of each 100 produced thereafter shall be selected at random. The sample shall have met the requirements and tests in Table XI and shall then meet the requirements and tests in Table XII. Unless otherwise specified in the contract, the items shall be returned to the lot upon successful completion of testing.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

TABLE XII. Special sampling.

Characteristic	Requirement	Test Procedures
Critical	None	N/A
Major 103. Vibration 104. Shock 105. Altitude 106. Temperature 107. Humidity (nonoperating)	 3.3.1 3.3.2 3.3.3 3.3.4 3.3.5	 4.7.1.1 4.7.1.2 4.7.1.3 4.7.1.4 4.7.1.5
Minor	None	N/A

4.5.1 Failure of sample. Should any one item of a special sampling fail to meet any of the specified test requirements, acceptance of the product shall be suspended by the Government until necessary corrections to all production since the last test, as well as to current production has been accomplished in accordance with the provisions of the contract.

4.6 Inspection equipment.

4.6.1 Government-furnished inspection equipment. Where the contract provides for Government-furnished test equipment, supply and maintenance of test equipment shall be in accordance with the applicable requirements of MIL-I-45607. The requirements of MIL-I-45607 shall not apply to Government-furnished TPSs. Supply and maintenance of Government-furnished TPSs shall be specified in the contract.

4.6.2 Contractor-furnished inspection equipment.

4.6.2.1 Government design. Unless otherwise specified in the contract, all inspection equipment identified by drawing number in specifications or QAPs forming a part of the contract shall be supplied by the contractor in accordance with design specified. The contractor may, however, propose alternatives to Government designs for approval in accordance with 6.3.

4.6.2.2 Contractor design. The contractor shall design and supply inspection equipment compatible with the "Methods of Inspection" specified in 4.7 of this specification and with the component inspection procedures specified in "Examination" and "Test Facilities" requirements of MIL-F-13926 wherever Government

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

designs are not specified (see 4.6.2.1). Since tolerance of test equipment is normally considered to be within 10 percent of the product tolerance for which it is intended, this inherent error in the test equipment design must be considered as part of the prescribed product tolerance limit. Thus, concept, construction, materials, dimensions, and tolerances used in the design of test equipment shall be so selected and controlled as to ensure that the test equipment will reliably indicate acceptability of a product which does not exceed 90 percent of the prescribed limit and permit positive rejection when nonconforming. Construction shall be such as to facilitate routine calibration of test equipment. Contractor inspection equipment designs shall be submitted for approval in accordance with 6.3.

4.6.2.3 Test equipment. In conjunction with 4.6.2.1 and 4.6.2.2, the standard test equipment listed in Table XIII (Test equipment list), or equivalent, shall be utilized in the performance of the applicable test methods of inspection as specified in 4.7. These items, which constitute the FCP's Test Program Set (TPS), or their equivalent, shall be used to test the FCP. Proper operation of the test equipment shall be verified before testing of the FCP is performed.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

TABLE XIII. Test equipment list.

Item Number	Description
1. Interface Device	Adapter E, PN 9360965
2. Cable Assembly	W1, PN 12563264
3. Cable Assembly	W2, PN 12563265
4. Cable Assembly	W3, PN 12563266
5. Cable Assembly	W4, PN 12563267
6. Cable Assembly	W5, PN 12563268
7. Cable Assembly	W6, PN 12563269
8. Cable Assembly	W7, PN 12563270
9. Cable Assembly	W8, PN 12563271
10. Cable Assembly	W9, PN 12563272
11. Digital Card Tested	AN/USM-465A
12. Comcode Program Tape	Version 2.0
13. Test Program Set Manual	TPS 12554988
14. Test Program Set Tape	TPT 12554988
15. Isolator (4 ea)	PN 10545465

4.7 Methods of inspection.

4.7.1 Environmental. Environmental tests shall be conducted in accordance with the test conditions, general test performance guidance, and individual test methods of MIL-STD-810, except as specified herein. Test facilities, chambers and apparatus used in conducting the tests contained in this specification shall be capable of meeting the conditions required.

4.7.1.1 Vibration. This test shall be performed as specified in Method 514, Category 8, Procedure I of MIL-STD-810 and paragraph 3.3.1 herein. The FCP shall be attached to the

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

vibration generator directly or with a fixture, and securely held by its normal means of attachment. After vibration, the FCP shall be tested as specified in 4.7.3 herein to determine compliance with 3.3.1.

4.7.1.2 Shock. This test shall be performed using the shock isolator listed in Table XIII and as specified in Procedure I, Section II of Method 516 of MIL-STD-810 except that:

a. The post-test functional check (Step 5 of Procedure I) need not be conducted until completion of all shock impulses.

b. The shock pulses shall be 15 g's ± 10 percent amplitude and 11 ms ± 10 percent duration.

c. The maximum g shall occur at 5.5 milliseconds ± 10 percent.

d. The shock measuring system shall have a flat frequency response within ± 10 percent from 20 to 2,000 Hertz.

After the application of all the shock impulses, the FCP shall be tested as specified in 4.7.3 to determine compliance with 3.3.2.

4.7.1.3 Altitude operating. This test shall be performed as specified in Procedures II, Method 500 of MIL-STD-810. The operational checkout shall be as specified in 4.7.3 for compliance with 3.3.3.

4.7.1.4 Temperature.

4.7.1.4.1 High (operating). This test shall be performed as specified in Procedures I and II, Method 501 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.3 for compliance with 3.3.4.

4.7.1.4.2 Low (operating and non-operating). This test shall be performed as specified in Procedures I and II, Method 502 of MIL-STD-810 with constant temperature extremes as specified in 3.3.4. The operational checkout shall be as specified in 4.7.3 for compliance with 3.3.4.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the current version before use.

4.7.1.5 Humidity (non-operating). This test shall be performed in accordance with Procedure III, Method 507 of MIL-STD-810 except that in Step 7, checkout shall be performed at the conclusion of the test only, with the item at standard ambient conditions. The operational checkout shall be as specified in 4.7.3 for compliance with 3.3.5.

4.7.2 Electrical. Continuity shall be inspected by standard measuring equipment and visual examination for compliance with 3.4.

4.7.3 Performance. The performance of the Fire Control Processor shall be tested to meet the requirements of 3.5, using the test equipment specified in 4.6. The FCP TPS shall test the requirements of Section 3.5 as specified in the following paragraphs. the TPS consists of 15 software modules executed consecutively. Power is initially applied to the FCP in module 7, causing the FCP embedded Start-Up BIT (SUBI) software to run. In module 8, the SUBI results are obtained. Modules 9 (E910) through 15 (E916) are FCP embedded self-tests run following Start-Up BIT if certain initial conditions are input by the TPS and exist simultaneously. These initial conditions are:

a. Power is applied to the FCP per 3.5.1.1.

b. $\overline{\text{RUN FCP UNIT TEST}}$ (DTFUT), J3-CC, is logic low (identified by SUBI).

c. Function Switch Position equals "C" hexadecimal (identified by SUBI).

If the Multiple Failure Message Counter (MFMC) equals 0, i.e., no faults are stored (as identified by module 9 at its beginning), module 9 shall generate and enter "E910" into the Off-Line BIT database pointer, then begin the embedded FCP self tests. Simulated weapon system voltages, input by the TPS for SUBI and modules 9 through 15, shall be per Table XIV.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

TABLE XIV. Fire Control Processor TPS inputs.

Input Parameter	Paragraph	Module							
		SUBI	9	10	11	12	13	14	15
DISPLAY/ENTRY	3.5.1.3.2.2 4.7.3.3.2	0	0	0	0	0	0	0	0
FUNCTION SWITCH	3.5.1.3.2.2 4.7.3.3.2.2	C	C	C	C	C	C	C	C
GROUND MODE	3.5.1.5.1.1, 2.1 4.7.3.5.1.1, 2.1	0	0	0	0	1*	0	0	0
RADAR MODE	3.5.1.5.1.2, 2.2 4.7.3.5.1.2, 2.2	0	0	0	0	0	1*	0	0
MANUAL MODE	3.5.1.5.1.3, 2.3 4.7.3.5.1.3, 2.3	0	0	1*	0	0	0	0	0
EXTERNAL MODE	3.5.1.5.1.4, 2.4 4.7.3.5.1.4, 2.4	0	0	0	1*	0	0	0	0
TEST MODE	3.5.1.5.1.5, 2.5 4.7.3.5.1.5, 2.5	1*	1*	0	0	0	0	1*	1*
ACTION SWITCH	3.5.1.5.1.6, 2.6 4.7.3.5.1.6, 2.6	0	0	0	0	0	0	0	1*
ACQ/TRACK	3.5.1.5.1.7, 2.7 4.7.3.5.1.7, 2.7	0	0	1	X	X	X	X	1*
FAULT BYPASS	3.5.1.5.1.8, 2.8 4.7.3.5.1.8, 2.8	0	0	1*	0	1*	1*	1*	0
FIRE TRIG RELAY	3.5.1.5.1.9, 2.9 4.7.3.5.1.9, 2.9	0	0	1*	X	X	X	X	0
EXT RNG ENABLE	3.5.1.5.1.10, 2.10 4.7.3.5.1.10, 2.10	1*	1*	X	0	X	X	X	1*
RADIATE COMMAND	3.5.1.5.1.11, 2.11 4.7.3.5.1.11, 2.11	1*	1*	0	X	X	X	X	1*
RUN DTOT	3.5.1.6.1.1, 2.1 4.7.3.6.1.1, 2.1	1	1	0	X	X	X	X	1
RUN TRAINING	3.5.1.6.1.1, 2.2 4.7.3.5.1.2, 2.2	1	1	0	X	X	X	X	1
RUN LEC LINK	3.5.1.6.1.3, 2.3 4.7.3.6.1.3, 2.3	1	1	0	X	X	X	X	1
RUN FCP UNIT TEST	3.5.1.6.1.4, 2.4 4.7.3.6.1.4, 2.4	0 0	0 0	1 1	X X	X X	X X	X X	1 1
SP/TWD	3.5.1.6.1.5, 2.5 4.7.3.6.1.5, 2.5	1	1	0	0	1	X	X	1
HAND CONTROL AZ	3.5.1.8.1 4.7.3.8.1.1	-9.5	9.5	0.0	0.0	0.0	0.0	0.0	-9.5
HAND CONTROL EL	3.5.1.8.1 4.7.3.8.1.2	-9.5	-9.5	0.0	0.0	0.0	0.0	0.0	9.5

(Refer to Page 46 for NOTES.)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the current version before use.

TABLE XIV. Fire Control Processor TPS inputs. (Continued)

Input Parameter	Paragraph	Module							
		SUBI	9	10	11	12	13	14	15
MANUAL RANGE	3.5.1.8.1 4.7.3.8.1.3	4.79	4.79	0.0	4.79	0.0	0.0	0.0	4.79
GUN TACH AZ	3.5.1.8.1 4.7.3.8.1.4	0.0	0.0	0.0	15	-15	15	-15	0.0
GUN TACH EL	3.5.1.8.1 4.7.3.8.1.5	0.0	0.0	0.0	15	-15	15	-15	0.0
CURRENT AZ 1	3.5.1.8.1 4.7.3.8.1.6	0	0	EAT	0	0	0	0	0
CURRENT AZ 2	3.5.1.8.1 4.7.3.8.1.7	0	0	EAT	0	0	0	0	0
CURRENT EL	3.5.1.8.1 4.7.3.8.1.8	0	0	EAT	0	0	0	0	0
MANUAL SPEED	3.5.1.8.2 4.7.3.8.2.1	1.68	1.68	0.0	0.0	0.0	0.0	0.0	1.68
AZ MOT 1 CW	3.5.1.8.2 4.7.3.8.2.2	0	0	EAT	0	0	0	0	0
AZ MOT 1 CCW	3.5.1.8.2 4.7.3.8.2.3	0	0	EAT	0	0	0	0	0
AZ MOT 2 CW	3.5.1.8.2 4.7.3.8.2.4	0	0	EAT	0	0	0	0	0
AZ MOT 2 CCW	3.5.1.8.2 4.7.3.8.2.5	0	0	EAT	0	0	0	0	0
EL MOT UP	3.5.1.8.2 4.7.3.8.2.6	0	0	EAT	0	0	0	0	0
EL MOT DOWN	3.5.1.8.2 4.7.3.8.2.7	0	0	EAT	0	0	0	0	0
DOWN LIMIT	3.5.1.10.1 4.7.3.10.1	1*	1*	0	0	1*	1*	0	1
UP LIMIT	3.5.1.10.2 4.7.3.10.2	0	0	1*	1*	0	0	1*	0
CCW LIMIT	3.5.1.10.3 4.7.3.10.3	1*	1*	0	0	1*	1*	0	1*
CW LIMIT	3.5.1.10.4 4.7.3.10.4	0	0	1*	1*	0	0	1*	0
SIGHT SYNCHRO AZ	3.5.1.11 4.7.3.11.1.1	0	0	EAT	EAT	0	0	0	0
SIGHT SYNCHRO EL	3.5.1.11 4.7.3.11.1.2	0	0	EAT	EAT	0	0	0	0
GUN SYNCHRO AZ	3.5.1.11 4.7.3.11.1.3	0	0	EAT	EAT	0	0	0	0
GUN SYNCHRO EL	3.5.1.11 4.7.3.11.1.4	0	0	EAT	EAT	0	0	0	0

(Refer to Page 46 for NOTES.)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

NOTES:

1. "1" means logic high (+5 Vdc)
   "1*" means logic high (+16 Vdc)

2. "0" means logic low (0 Vdc)

3. "X" means logic level is a "don't care"

4. "C" is hexadecimal

5. "EAT" means input is from an output

4.7.3.1 Power input/output.

4.7.3.1.1 Power input. Run FCP TPS modules 7, 8, and 9 to apply the input voltages of Table XV to the FCP connector-pins shown and verify the inputs remain within the tolerance shown. If any input is out of tolerance, either a failure message or a fault code shall be printed out as noted.

TABLE XV. Power inputs.

Input Voltage	Connector-Pins	Measured Input and Tolerance (Msg/Code)		
		Module 7	Module 8	Module 9
115 VAC 400 Hz	J2-M (R1) J2-K (R2)	115 +25/-55 Vrms (Msg)	400.0 ± 0.4 Hz (F082)	--
+5 Vdc	J2-D,E (+) J2-P,F RTN	+5 +2/-1 Vdc (Msg)	+5.00 ± 0.25 Vdc (F033)	--
+15 Vdc	J2-A (+) J2-N (RTN)	+15 +6/-3 Vdc & +15 ± 6 Vdc (Msg)	+15.00 ± 0.75 Vdc (F037)	--
-15 Vdc	J2-B (-) J2-N (RTN)	-15 +4/-8 Vdc & -15 ± 8 Vdc (Msg)	-15.00 ± 0.75 Vdc (F038)	--
+24 Vdc	J2-H (+) J2-R (RTN)	+24 +8.5/-5.5 Vdc (Msg)	--	+24.0 +1.2/ -3.0 Vdc (F703)
-24 Vdc	J2-J (-) J2-R (RTN)	-24 +2/-14.5 Vdc (Msg)	--	-24.0 -1.2/ +3.0 Vdc (F704)
+24 Vdc BAT	J1-(R),(C), W (+) J1-R (RTN)	+24 +8.5/-5.5 Vdc (Msg)	--	--

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the current version before use.

4.7.3.1.2 Power output. Run FCP TPS module 7 to verify the voltages of Table XVI are output from the FCP within the tolerances shown. If any output is out of tolerance, a failure message noting the failed output shall be printed.

TABLE XVI. Power outputs.

Voltage Output	Connector-Pin	Measured Output and Tolerance
115 Vac 400 Hz	J4-J (R1) J4-K (R2) J7-X (R1) J7-Y (R2)	115 ± 38 Vac
+5 Vdc	J7-(R),(K) J3-GG,HH RTN J7-(M),(S) RTN	+5 +1/-2 Vdc
+15 Vdc	J1-(F) J1-U RTN	+15 +3/-6 Vdc
-15 Vdc	J1-M J1-U RTN	-15 +6/-3 Vdc
+24 Vdc	J4-(R) J4-(M) RTN	+24 +3.8/-10.1 Vdc
-24 Vdc	J4-(N) J4-(M) RTN	-24 +8.9/-6.2 Vdc
+24 Vdc BAT	J4-R J3-X RTN J6-(B),(R) RTN	+24 +3.8/-10.1 Vdc

4.7.3.2 Fan. Run FCP TPS module 8 (Start-Up BIT) to verify the fan speed is 6500 ±3050 rpm. If the fan speed is out of tolerance, fault message F701 shall be output.

4.7.3.3 Serial data communication interfaces.

4.7.3.3.1 Data clock outputs. Run FCP TPS module 8 to verify the data clock outputs of paragraph 3.5.1.3.1 are transitioning from logic high to logic low at a greater than 200 kHz rate. If the clock outputs are not transitioning at a rate greater than 200 kHz, a fault message shall be printed out.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

4.7.3.3.2 Control panel communications. Run FCP TPS module 8 (Start-Up BIT) to verify the FCP can transmit and receive data over an external loop; i.e., perform an end-around-test (EAT) through Interface Device E with correct timing, status bits and error free data. If timing, status bits or data are incorrect, fault F022 shall be output.

4.7.3.3.2.1 Transmitting. Run FCP TPS module 8 to transmit data out port CPU TO CONTROL PANEL DATA at J7-T (HI) and J7-U (LO) to request function switch position, to initiate an end-around-test, and to output the next setup message (Transmitting a setup message exercises 4 bits of the transmitted 8-bit data byte.) Run TPS module 9 to exercise the Display/Entry and Enter switch bits of the 8-bit data byte. Run TPS module 10 to verify the IOB control panel UART's TBRE and TRE status bits both go logic low when two bytes are transmitted, otherwise F022 shall be output.

4.7.3.3.2.2 Receiving. Run FCP TPS module 8 to receive hexadecimal "C" from interface device E through port CONTROL PANEL TO CPU DATA at J7-P (HI) and J7-N (LO) for initiating Start-Up BIT. If hexadecimal "C" is not received, a fault message to that effect shall be printed.

4.7.3.3.3 Distribution Box, Unit2 and Unit 5 communications.

4.7.3.3.3.1 Distribution Box communications. Run FCP TPS module 9 to do an internal EAT of the IOB's BITE UART for checking data timing and accuracy. Module 9 shall check the BITE UART's Data Ready (DR) bit status, using an open loop test, and ensure it does not set (because the transmitted data should not come back). Module 9 shall output F022 for data and timing faults (internal EAT loop) and F133 for DR logic faults (open loop). Run FCP TPS module 10 to transmit data out port CPU TO DISTRIBUTION BOX DATA at J8-S (HI) and J8-T (LO) and receive data into port DISTRIBUTION BOX TO CPU DATA at J8-U (HI) and J8-V (LO). Module 10 shall check the logic and timing of bits DR, TBRE and TRE and output F022 if a fault exists. Module 10 shall also check the received data against the transmitted data and output F133 if a fault exists.

4.7.3.3.3.2 Unit 2 communications. Run FCP TPS module 10 to transmit data out port CPU TO UNIT 2 DATA at J7-J (HI) and J7-K (LO) and receive data into port UNIT 2 TO CPU DATA at J7-L (HI) and J7-M (LO). (This is an external EAT through Interface Device E.) This test checks data timing and accuracy and the Data Ready (DR) status bit for correct logic level. If a fault exists, F134 shall be output.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the current version before use.

4.7.3.3.3.3 Unit 5 communications. Run FCP TPS module 10 to transmit data out port CPU TO UNIT 5 DATA at J6-D (HI) and J6-U (LO) and receive data into port UNIT 5 TO CPU DATA at J6-V (HI) and J6-W (LO). (This is an external EAT through Interface Device E.) This test shall check data timing and accuracy and the DR status bit for correct logic level. If a fault exists, F135 shall be output.

4.7.3.3.4 Unit 4 communications. Run FCP TPS module 9 to do an internal EAT on the IOB's Unit 4 UART. These tests shall check the order in which the low and high bytes are reassembled upon receipt to form a 16-bit word. If the order is incorrect, either F022 or F136 shall be output. Two data words are sent to toggle all bits and ensure none are stuck logic high (SA1) or logic low (SA0). If the bits are incorrect, F022 shall be output, Run FCP TPS module 10 to transmit data out port CPU TO UNIT 4 DATA at J6-N (HI) and J6-M (LO) and receive data into port UNIT 4 TO CPU DATA at J6-L (HI) and J6-(D) (LO). (This is an external EAT through Interface Device E.) This test shall check data byte sequencing for word formation and, if incorrect, output F136. IOB Unit 4 UART status bits TBRE and TRE shall also be checked for logic level and timing and, if incorrect, F022 shall be output.

4.7.3.3.5 Link #1 communications. Run TPS module 9 to do an internal EAT to check First-In First-Out (FIFO) shift register sequencing and timing (37 milliseconds max.), using 128 words, and to check operation of the high and low byte UARTs. All data lines shall be toggled to ensure none are stuck logic high or low. Logic levels and timing of the UART Data Ready signals are also be checked. Fault F605 shall be output if any of the above items fail. Run TPS module 10 to verify the high and low byte Transmitter Buffer Register Empty and Transmitter Register Empty lines of the UARTs are not stuck logic high or low. F613 shall be output if any of these lines are stuck logic high or low. Module 10 also checks the RUN DTOT and RUN TRAINING inputs (J3-AA and J3-BB, respectively) for logic lows and shall output F603 if not low. Run module 11 to do an external EAT through Interface Device "E", using 89AB hexadecimal as data input to the FIFOs. The data is transmitted out ports CPU TO DATA LINK #1 HIGH DATA at J3-C (HI), J3-D (LO) and CPU TO DATA LINK#1 TO CPU HI DATA at J3-(F) (HI), J3-(H) (LO) and DATA LINK #1 TO CPU LOW DATA at J3-(I) (HI), J3-(J) (LO). Fault F613 shall be output if the returned data does not match the transmitted data.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

4.7.3.3.6 Link #2 and Line #3 communications.

4.7.3.3.6.1 Link #2. Run TPS module 6 to check RUN LEC LINK (J3-EE) for stuck logic low. If J3-EE is SAO, the TPS shall remove power from the TPS. Run TPS module 10 to check RUN LEC LINK for stuck logic high and output F611 if it is. Module 10 shall also check the UART TBRE and TRE lines for stuck logic high (F614) and stuck logic low (F022). Transmitter timing is checked by determining if TBRE is logic high and TRE is logic low after the first byte of a two byte load has been transmitted (0.65 ± 0.01 milliseconds into a word transmit cycle). F614 shall be output if logic levels are incorrect.

4.7.3.3.6.2 Link #3. Run TPS module 6 to check $\overline{\text{RUN SPARE}}$ (J3-DD) for stuck logic low, and if so, FCP power shall be turned off by the TPS. (J3-DD is not connected in the present production version of the FCP.) Run E910 to do a Link #2/Link #3 UART EAT by transmitting and receiving bytes AA and 55 hexadecimal and to check DR logic levels and timing. For any error, F606 shall be output. Run E911 to check WSTBRE and WSTRE for logic highs with no data being transmitted, and if both are not logic high, F614 shall be output.

4.7.3.4 Continuity.

a. Run TPS module 5 to pull FOOT SWITCH (J5-B) logic high and check for logic highs at ACTION SWITCH RELAY (J1-(K)) and ANTENNA ACTION SWITCH (J6-(G)). A SAO fault message shall be output if the test fails. Module 5 also repeats the test with J5-B pulled logic low and checks J1-(K) and J6-(G) for logic lows. A SA1 fault message shall be output if the test fails.

b. Run TPS module 7 to check EXTERNAL MODE (J1-(J)) for stuck logic high (+24 V) with J1-(M) open circuited. A SA1 fault message shall be output if the test fails. Run TPS module 11 to check EXTERNAL MODE J1-(J) for stuck logic low with +24 Vdc applied to J1-(M). F003 shall be output if the test fails.

c. Run TPS module 5 to pull RADAR OPERATE (J1-X) logic high and check for a logic high at RADAR OPERATE (J6-Y). A SA0 fault message shall be output if the test fails. Module 5 also pulls J1-X logic low and checks for a logic low at J6-Y. A SA1 fault message shall be output if the test fails.

d. Run TPS module 10 to pull FIRE TRIGGER RELAY (J1-N) logic high and check for a logic high at FIRE TRIGGER RELAY (J6-X). A SA0 fault message shall be output if the test fails.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the current version before use.

e. Run TPS module 7 to pull FOOT SWITCH (J5-A) logic high and check for logic high at RADIATE COMMAND (J6-(H)). A SA0 fault message shall be output if the test fails. Run TPS module 10 to pull J5-A logic low and check for logic low at J6-(H). A SA1 fault message shall be output if the test fails.

4.7.3.5 High level discrete input signals.

4.7.3.5.1 Logic high (+16 ± 1 Vdc).

4.7.3.5.1.1 GROUND MODE. Run TPS module 12 to apply +16 Vdc to J1-(P) and compare the IOB's mode word on the data bus (VMSPCP) to the constant for GROUND MODE asserted (#GROUND). If they are equal, the mode word on the SCB (SCMW) is compared to the constant for ground mode on the SCB (#GMB). If not equal, the SCB fails and F008 shall be output. If the IOB's mode word fails and the SCB's passes, F012 shall be output. If both fail, F013 shall be output.

4.7.3.5.1.2 RADAR MODE. Run TPS module 13 to apply +16 Vdc to J1-(B) and compare the IOB's mode word on the data bus (VMSPCP) to the constant for RADAR MODE asserted (#RADAR). If not equal, F003 shall be output.

4.7.3.5.1.3 MANUAL MODE. Run TPS module 10 to apply +16 Vdc to J1-E and compare the IOB's mode word on the data bus (VMSPCP) to the constant for MANUAL MODE asserted (#MANUAL). If not equal, F003 shall be output.

4.7.3.5.1.4 EXTERNAL MODE. Run TPS module 11 to apply +16 Vdc to J1-(M) and compare the IOB's mode word on the data bus (VMSPCP) to the constant for EXTERNAL MODE asserted (#EXTERNAL). If not equal, F003 shall be output.

4.7.3.5.1.5 TEST MODE. Run TPS module 9 to apply +16 Vdc to J1-(D) and compare the IOB's mode word on the data bus (VMSPCP) to the constant for TEST MODE asserted (#TESTPOS). If not equal, F003 shall be output.

4.7.3.5.1.6 ACTION SWITCH. Run TPS module 15 to apply +16 Vdc to J1-A. If the action switch bit on the IOB (ACT) is logic high and the action switch bit on the SCB (ASB) is logic low, the test shall pass. If ACT was logic low and ASB was logic low, F005 shall be output. If ACT was logic low and ASB was logic high, F021 shall be output. If ACT was logic high and ASB was logic high, F007 shall be output. Module 15 also checks for activation of the DTB's action switch relay on bit D5 (RSAS) of the relay status word (DTRELYST). If bit RSAS is logic low, this test passes, and if logic high, this test fails and F604 shall be output.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

4.7.3.5.1.7 ACQUISITION/TRACK SWITCH. Run TPS module 10 to apply 16 Vdc to J1-(S). If the ACQ/TRK bit on the IOB (AKTRK) is logic high and the DTB's relay status bit (RSATS) in the relay status word (DTRELYST) is logic low, the test passes. If AKTRK was logic low, F004 shall be output. If AKTRK was logic high and RSATS was logic high, F604 shall be output.

4.7.3.5.1.8 FAULT BYPASS. Run TPS module 10 to apply +16 Vdc to J1-Z. If the fault bypass bit on the IOB (FBS) is logic high and the fault bypass bit on the SCB (FBB) is logic low, the test passes. If FBS is logic high and FBB is logic high, F009 shall be output. If FBS is logic low and FBB is logic low, F010 shall be output. If FBS is logic low and FBB is logic high, F011 shall be output.

4.7.3.5.1.9 FIRE TRIGGER RELAY. Run TPS module 10 to apply 16 Vdc to J1-N. If the gun fire trigger bit on the IOB (GFTX) is logic high and the DTB's relay status bit (RSFTR) in the relay status word (DTRELYST) is logic low, the test passes. If GFTX is logic low, F142 shall be output. If GFTX is logic high and RSFTR is logic high, F604 shall be output.

4.7.3.5.1.10 EXTERNAL RANGE ENABLE. Run TPS module 9 to apply +16 Vdc to J1-(G). If the external range enable bit on the IOB (ERE) is logic high, the test passes. If ERE is logic low, F120 shall be output.

4.7.3.5.1.11 FOOT SWITCH. Run TPS module 9 to apply +16 Vdc to J5-A. If the Radiate Command bit (RADC) on the IOB is logic high, the test passes. If RADC is logic low, F124 shall be output. Module 9 also checks the Radiate Command relay on the DTB. If bit RSRC of the relay status word DTRELYST is logic low, the test passes; if logic high, F604 shall be output.

4.7.3.5.2 Logic low (0 ± 1 Vdc).

4.7.3.5.2.1 GROUND MODE. Run TPS module 9 to apply 0 Vdc to J1-(P). If the Ground Mode bit at the SCB (GMB) is logic high, the test passes. If GMB is logic low, F008 shall be output.

4.7.3.5.2.2 RADAR MODE. RADAR MODE is not tested logic low by the TPS.

4.7.3.5.2.3 MANUAL MODE. MANUAL MODE is not tested logic low by the TPS.

4.7.3.5.2.4 EXTERNAL MODE. EXTERNAL MODE is not tested logic low by the TPS.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

4.7.3.5.2.5 TEST MODE. TEST MODE is not tested logic low by the TPS.

4.7.3.5.2.6 ACTION SWITCH. Run TPS module 10 to apply 0 Vdc to J1-A. If the Action Switch bit on the IOB (ACT) is logic low and the Action Switch bit on the SCB (ASB) is logic high, the test passes. If ACT is logic high and ASB is logic high F005, shall be output. If ACT is logic high and ASB is logic low, F006 shall be output. If ACT is logic low and ASB is logic low, F007 shall be output. If the above tests of the Action Switch passed, module 10 also checks the Action Switch relay bit (RSAS) of relay status word DTRELYST from the DTB. If RSAS is logic high, the test passes. If RSAS is logic low, ACT is read again. If ACT is now logic high, F604 shall be output, and if logic low F077 shall be output. Run module 15 to check RSAS for a logic low with ACT logic high and ASB logic low. F604 shall be output if RSAS is logic high.

4.7.3.5.2.7 ACQUISITION/TRACK SWITCH. Run TPS module 9 to input 0 Vdc to J1-(S). If the Acquisition/Track bit at the IOB (AKTRK) is logic low, the test passes. If AKTRK is logic high, F004 shall be output. Module 9 also checks the Acquisition/Track relay on the DTB. If the RSATS bit of relay status word DTRELYST is logic high, the test passes. If RSATS is logic low, AKTRK is read again. If AKTRK is now logic low, F604 shall be output, and if logic high, F077 shall be output.

4.7.3.5.2.8 FAULT BYPASS. Run TPS module 15 to input 0 Vdc to J1-Z. If the Fault Bypass bit on the IOB (FBS) is logic low and the Fault Bypass bit on the SCB (FBB) is logic high, the test passes. If FBS is logic low and FBB is logic low, F009 shall be output. If FBS is logic high and FBB is logic low, F011 shall be output.

4.7.3.5.2.9 FIRE TRIGGER RELAY. Run TPS module 9 to apply 0 Vdc to J1-N. If the Fire Trigger Relay bit on the IOB (GFTX) is logic low, the test passes. If GFTX is logic high, F142 shall be output. Module 9 also checks the Fire Trigger Relay relay on the DTB. If the RSFTR bit of relay status word DTRELYST is logic high, the test passes. If RSFTR is logic low, GFTX shall be read again. If GFTX is logic low, F604 shall be output, and if logic high, F077 shall be output.

4.7.3.5.2.10 EXTERNAL RANGE ENABLE. Run TPS module 11 to apply 0 Vdc to J1-(G). If the External Range Enable bit on the IOB (ERE) is logic low the test passes. If ERE is logic high, F120 shall be output.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

4.7.3.5.2.11 FOOT SWITCH. Run TPS module 10 to apply 0 Vdc to J5-A. If the Radiate Command bit on the IOB (RADC) is logic low, the test passes. If RADC is logic high, F124 shall be output.

4.7.3.6 Low level (ATTL) discrete input signals.

4.7.3.6.1 Logic high (+5 Vdc).

4.7.3.6.1.1 RUN DTOT. Run TPS module 9 to apply +5 Vdc to J3-AA. If the RUN DTOT bit (DTLNKON) of DTB status word DTTSTAT is logic high, the test passes. If DTLNKON is logic low, F603 shall be output.

4.7.3.6.1.2 RUN TRAINING. Run TPS module 9 to apply +5 Vdc to J3-BB. If the RUN TRAINING bit (DTTRNON) of DTB status word DTTSTAT is logic high, the test passes. If DTTRNON is logic low, F603 shall be output.

4.7.3.6.1.3 RUN LEC LINK. Run TPS module 6 to apply +5 Vdc to J3-EE. FCP power shall be turned off by the TPS if J3-EE is SA0.

4.7.3.6.1.4 RUN FCP UNIT TEST. Run TPS module 6 to apply +5 Vdc to J3-CC. FCP power shall be turned off by the TPS if J3-CC is SA0. Also run TPS module 10 to apply +5 Vdc to J3-CC. If the RUN FCP UNIT TEST bit (DTFUT) of DTB status word DTTSTAT is logic high, the test passes. If DTFUT is logic low, F612 shall be output.

4.7.3.6.1.5 SP/TWD. Run TPS module 9 to apply +5 Vdc to J8-GG. If the SP/TWD bit (TOWSP) of the Servo Controller Mode Word (SCMW) is logic high, the test passes. If TOWSP is logic low, F710 shall be output.

4.7.3.6.2 Logic low (0 Vdc).

4.7.3.6.2.1 RUN DTOT. Run TPS module 10 to apply 0 Vdc to J3-AA. If the RUN DTOT bit (DTLNKON) of DTB status word DTTSTAT is logic low, the test passes. If DTLNKON is logic high, F603 shall be output.

4.7.3.6.2.2 RUN TRAINING. Run TPS module 10 to apply 0 Vdc to J3-BB. If the RUN TRAINING bit (DTTRON) of DTB status word DTTSTAT is logic low, the test passes. If DTTRNON is logic high, F603 shall be output.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the currentversion before use.

4.7.3.6.2.3 RUN LEC LINK. Run TPS module 10 to apply 0 Vdc to J3-EE. If the RUN LEC LINK bit (WSLO) of DTB status word LNKWSTAT is logic low, the test passes. If WSLO is logic high, F611 shall be output.

4.7.3.6.2.4 RUN FCP UNIT TEST. Run TPS module 8 to apply 0 Vdc to J3-CC. If the RUN FCP UNIT TEST bit (DTFUT) of DTB status word DTTSTAT is logic low, E910 (module 9) shall begin (if all other conditions are satisfied). If DTFUT is logic high, the E910 through E916 setups (modules 9 through 15) of the FCP TPS shall not run.

4.7.3.6.2.5 SP/TWD. Run TPS module 10 to apply 0 Vdc to J8-GG. If the SP/TWD bit (TOWSP) of the Servo Controller Mode Word (SCMW) is logic low, the test passes. If TOWSP is logic high, F710 shall be output.

4.7.3.7 Discrete output signals.

4.7.3.7.1 COMPUTER POWER LAMP. Run TPS module 8 to check J1-K for a logic low with FCP power applied. A SA1 fault message shall be output if J1-K is logic high.

4.7.3.7.2 COMPUTER CYCLE. Run TPS module 9 to cause the COMPUTER CYCLE (also known as COMPUTER GOOD) output at J1-V to transition logic low at a 4 ± 0.5 Hertz rate. The AN/USM-465 monitors this output and shall print a message stating the test passes or J1-V is SA0 or SA1.

4.7.3.7.3 ACTION SWITCH relay. Run TPS module 5 to check normally open relay contact J3-Z for a logic high with the relay de-energized. If J3-Z is logic high, the test passes. If J3-Z is logic low, a SA0 fault message shall be printed.

4.7.3.7.4 ACTION SWITCH LOCKOUT. Run TPS module 15 to set SALO logic high with no fault bypass switch input. ACTION SWITCH LOCKOUT output J1-(A) shall transition logic low to energize a relay to interrupt the +16 Vdc Action Switch input to J1-A within 0.25 ± 0.01 second. If this test fails, SALO is made logic low and bit D4 (SALOI) is tested to be logic low. If SALOI is logic low, F023 shall be output, and if logic high, F022 shall be output. Run TPS module 9 to set SALO logic high with no fault bypass switch input. SALOI should be logic high. If SALOI is logic low, F141 shall be output.

4.7.3.7.5 READY TO FIRE LAMP and relay. Run TPS module 9 to cause the READY TO FIRE LAMP output at J4-P to transition logic low at a 4 ± 0.5 Hertz rate. The AN/USM-465 tester shall monitor this output and print a message stating the test passed

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

or J4-P is SA0 or SA1. Normally Open relay contact J3-(B) is pulled logic high by a pull-up resistor and the Common relay contact, J3-(A) is grounded to +5 V RTN by TPS Interface Device "E". Run TPS module 5 to check J3-(B) for a logic high and print a SA0 fault message if J3-(B) is logic low. Run TPS module 9 to set the Ready to Fire Lamp bit, and check the status of the word containing the Ready To Fire relay contacts connected to the data bus, for a logic low. If the RSRTF bit of status word DTRELYST is logic high, F604 shall be output.

4.7.3.7.6 FAULT Lamp. Run TPS module 9 to cause the FAULT LAMP output at J1-(E) to transition logic low at a 4 ± 0.5 Hertz rate. The AN/USM-465 tester shall monitor this output and print a message stating the test passed or J1-(E) is SA0 or SA1.

4.7.3.7.7 TRACK/JAM LAMP and relay. Run TPS module 9 to cause the TRACK/JAM LAMP output at J4-S to transition logic low at 4 ± 0.5 Hertz rate. The AN/USM-465 tester shall monitor this output and print a message stating the test passed or J4-S is SA0 or SA1. Module 9 also sets the Track/Jam Radiate Bit (TJRB) to a logic high and verifies the Relay Status Track Jam (RSTJ) bit of the DTRELYST word is logic low. If the RSTJ bit is logic high, F604 shall be output. If the RSTJ bit is logic low, TJRB is made logic low and RSTJ is checked for a logic high. If RSTJ is logic high, the test passes; if low, F604 shall be output.

4.7.3.7.8 ACQ/TRACK SWITCH relay. Normally Closed relay contact J3-R and Normally Open relay contact J3-T are pulled logic high by pull-up resistors on TPS Interface Device "E" (IDE). The relay's Common contact is grounded to the +5 V RTN by the IDE. Run TPS module 5 to check Normally Closed contact J3-R for a logic low with the relay de-energized. If J3-R is logic low, the test passes. If J3-R is logic high, an "open connection between J3-R and J3-S" message shall be printed. Module 5 also checks J3-T for a logic high with the relay de-engergized and shall print a SA0 message if J3-T is logic low. Run TPS module 10 to check Normally Open contact J3-R for a logic high with the relay energized. A "J3-R SAO" message shall be printed if J3-R is logic low. Module 10 also checks J3-T for a logic low with the relay energized and if J3-T is logic low, the test passes. If J3-T is logic high, "J3-T SA1" shall be printed.

4.7.3.7.9 RADIATE COMMAND relay. Normally Open relay contact J3-V is pulled up to a logic high by Interface Device "E". The relay's Common contact is grounded to the +5 V RTN by the IDE. Run TPS module 5 to check Normally Open contact J3-V for a logic high with the relay de-energized. If J3-V is logic high, the test passes. If J3-V is logic low, fault message "J3-V SA0" shall be printed.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the current version before use.

4.7.3.7.10 FIRE TRIGGER RELAY relay. Normally closed relay contact J3-(G) and Normally Open relay contact J3-(E) are pulled up to logic highs by Interface Device "E". The relay's Common contact is grounded to the +5 V RTN by the IDE. Run TPS module 5 to check Normally Closed contact J3-(G) for a logic low with the relay de-energized. If J3-(G) is logic low, the test passes. If J3-(G) is logic high, fault message "Open connection between J3-(C) and (G)" shall be printed. Module 5 also checks Normally Open contact J3-(E) for a logic high with the relay de-energized. If J3-(E) is logic high, the test passes. If J3-(E) is logic low, fault message "J3-(E) SA0" shall be printed.

4.7.3.8 Analog input signals.

4.7.3.8.1 SCB scaled, selected and converted input signals.

4.7.3.8.1.1 HAND CONTROL AZ. Run TPS module 9 to input a HAND CONTROL AZ voltage of +9.5 ± 0.5 Vdc to J1-P (HI) and J1-F (LO). If the measured Sight Composite Rate in Azimuth (SCRA) from the SCP is less than -9.0 Vdc, the test passes. If SCRA is not less than -9.0 Vdc, F015 shall be output.

4.7.3.8.1.2 HAND CONTROL EL. Run TPS module 9 to input a HAND CONTROL EL voltage of -9.5 ± 0.47 Vdc to J1-D (HI) and J1-(N) (LO). If the measured Sight Composite Rate in Elevation (SCRE) is not greater than +9.0 Vdc, F016 shall be output.

4.7.3.8.1.3 MANUAL/EXTERNAL RANGE. Run TPS module 9 to input a target range voltage of +4.79 ±0.47 Vdc to MANUAL/-EXTERNAL RANGE input J1-(H)+ and J1-U RTN. If the measured Target Range Potentiometer Position (TRPP) is 1500 ± 150 meters, the test passes. If TRPP is not 1500 ± 150 meters, F036 shall be output.

4.7.3.8.1.4 GUN TACH AZ. Run TPS module 9 to input an Azimuth Turret Tach (ATT) voltage of 0.00 Vdc to J8-A (HI) and J8-B (LO) and measure an azimuth turret rate of less than 7.3 milliradians/second. If the absolute value of ATT is not less than 7.3 milliradians/second, F708 shall be output. Run TPS module 11 to input an azimuth tach voltage of +15.00 ± 0.75 Vdc and measure an azimuth turret rate of 184 ± 20 milliradians/-second. If ATT is not 184 ±20 milliradians/second, F708 shall be output. Run TPS module 12 to input an azimuth tach voltage of -15.00 ± 0.75 Vdc and measure an azimuth turret rate of -605.5 ± 40 milliradians/second. If ATT is not -605.5 ± 40 milli-radians/second, F708 shall be output.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the currentversion before use.

4.7.3.8.1.5 GUN TACH EL. Run TPS module 9 to input on Elevation Gun Tach (EGT) voltage of 0.00 Vdc to J8-(X) (HI) and J8-(Y) (LO) and measure an elevation gun rate of less than 39.4 milliradians/second. If the absolute value of EGT is not less than 39.4 milliradians/second, F709 shall be output. Run TPS module 11 to input an elevation tach voltage of +15.00 ± 0.75 Vdc and measure an elevation gun rate of 85.5 ± 10 milliradians/second. If EGT is not 85.5 ±10 milliradians/second, F709 shall be output. Run TPS module 12 to input an elevation tach voltage of -15.00 ± 0.75 Vdc and measure an elevation gun rate of 0256.7 ± 18 milliradians/second. If EGT is not -256.7 ± 18 milliradians/second, F709 shall be output.

4.7.3.8.1.6 CURRENT AZ 1. Run TPS module 9 to input 0.00 Vdc into CURRENT AZ 1 (AMC1) at J8-D (HI) and J-E (LO) to represent the output current from azimuth turret drive servo amplifier #1. If the absolute value of AMC1 is not less than 0.147 Adc, the test passes. If the absolute value of AMC1 is not less than 0.147 Adc, F711 shall be output. Run TPS module 10 to input +2.5 to +3.0 Vdc into CURRENT AZ 1 and measure AMC1. If AMC1 measures between 37.5 and 46.1 Adc, the test passes. If AMC1 does not measure between 37.5 and 46.1 Adc, F711 shall be output.

4.7.3.8.1.7 CURRENT AZ 2. Run TPS module 9 to input 0.00 Vdc into CURRENT AZ 2 (AMC2) at J8-J (HI) and J8-K (LO) to represent the output current from azimuth turret drive servo amplifier #2. If the absolute value of AMC2 is less than 0.147 Adc, the test passes. If the absolute value of AMC2 is not less than 0.147 Adc, F712 shall be output. Run TPS module 10 to input +2.5 to +3.0 Vdc into CURRENT AZ 2 and measure AMC2. If AMC2 measures between 37.5 and 46.1 Adc, the test passes. If AMC2 does not measure between 37.5 and 46.1 Adc, F712 shall be output.

4.7.3.8.1.8 CURRENT EL. Run TPS module 9 to input 0.00 Vdc into CURRENT EL (EMC) at J8-(D) (HI) and J8-(E) (LO) to represent the output current from the elevation gun servo amplifier. If the absolute value of EMC is less than 0.147 Adc, the test passes. If the absolute value of EMC is not less than 0.147 Adc, F713 shall be output. Run TPS module 10 to input +0.7 to +1.3 Vdc into CURRENT EL and measure EMC. If EMC measures between 12.6 and 14.9 Adc, the test passes. If EMC does not measure between 12.6 and 14.9 Adc, F713 shall be output.

4.7.3.8.1.9 COMPOSITE SCP RATE COMMANDS. Run TPS module 9 to input and sum the Sight Command Az, Sight Command El, Gun Command Az, and Gun Command El rate output signals. These signals are summed when all are set to zero (to obtain the zero

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the current version before use.

offset) and when individually set to 4000, 3FFF, C000, and C001 hexadecimal with the other outputs set to zero. If any averaged sum of seven samples minus the zero offset exceeds the allowable error, F031 shall be output.

4.7.3.8.2 IOB scaled, selected and converted input signals.

4.7.3.8.2.1 MANUAL SPEED. Run TPS module 9 to input a target speed voltage of 1.68 ± 0.15 Vdc to MANUAL SPEED input J1-T (=) and J1-U RTN. If the measured Target Speed Potentiometer Position (TSPP) is 300 ± 30 knots, the test passes. If TSPP is not 300 ± 30 knots, F035 shall be output.

4.7.3.8.2.2 AZ MOT 1 CW. The FCP TPS connects the AZ MOT 1 CW input at J8-AA, J8-HH (RTN) to the output of the Sight Command Az DAC through Interface Device "E", when the TPS commands the external EAT loop to close. Run TPS module 9 to leave the EAT loop open. The AZ MOT 1 CW input voltage will, therefore, be zero. The absolute value of the AZ MOT 1 CW input, after scaling, shall be less than 0.079 Vdc, otherwise F127 shall be output. Run TPS module 10 to combine an azimuth yoke command of 0 Vdc with a Sight Rate Azimuth Command of +10 Vdc, output the difference from the DAC, and close the EAT lop through the IDE. The scaled AZ MOT 1 CW voltage shall then be 1.42 ± 0.079 Vdc, otherwise F127 shall be output.

4.7.3.8.2.3 AZ MOT 1 CCW. The FCP TPS connects the AZ MOT 1 CCW input at J8-BB, J8-HH (RTN) to the output of the Sight Command El DAC through Interface Device "E", when the TPS commands the external EAT loop to close. Run TPS module 9 to leave the EAT loop open. The AZ MOT 1 CCW input voltage will, therefore, be zero. The absolute value of the AZ MOT 1 CCW input, after scaling, shall be less than 0.079 Vdc, otherwise F128 shall be output. Run TPS module 10 to combine an elevation yoke command of 0 Vdc with a Sight Rate Elevation Command of +10 Vdc, output the difference from the DAC, and close the EAT loop through the IDE. The scaled AZ MOT 1 CCW voltage shall then be 1.42 ± 0.079 Vdc, otherwise F128 shall be output.

4.7.3.8.2.4 AZ MOT 2 CW. The FCP TPS connects the AZ MOT 2 CW input at J8-CC, J8-HH (RTN) to the output of the Sight Command Az DAC through Interface Device "E", when the TPS commands the external EAT loop to close. Run TPS module 9 to leave the EAT loop open. The AZ MOT 2 CW input voltage will, therefore, be zero. The absolute value of the AZ MOT 2 CW input, after scaling, shall be less than 0.079 Vdc, otherwise F129 shall be

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

output. Run TPS module 10 to combine an azimuth yoke command of 0 Vdc with a Sight Rate Azimuth Command of +10 Vdc, output the difference from the DAC, and close the EAT loop through the IDE. The scaled AZ MOT 2 CW voltage shall then be 1.42 ± 0.079 Vdc, otherwise F129 shall be output.

4.7.3.8.2.5 AZ MOT 2 CCW. The FCP TPS connects the AZ MOT 2 CCW input at J8-DD, J8-HH (RTN) to the output of the Sight Command El DAC through Interface Device "E", when the TPS commands the external EAT loop to close. Run TPS module 9 to leave the EAT loop open. The AZ MOT 2 CCW input voltage will, therefore, be zero. The absolute value of the AZ MOT 2 CCW input, after scaling, shall be less than 0.079 Vdc, otherwise F130 shall be output. Run TPS module 10 to combine an elevation yoke command of 0 Vdc with a Sight Rate Elevation Command of +10 Vdc, output the difference from the DAC, and close the EAT loop through the IDE. The scaled AZ MOT 2 CCW voltage shall then be 1.42 ± 0.079 Vdc, otherwise F130 shall be output.

4.7.3.8.2.6 EL MOT UP. The FCP TPS connects the EL MOT UP input at J8-EE, J8-HH (RTN) to the output of the Sight Command Az DAC through Interface Device "E", when the TPS commands the external EAT loop to close. Run TPS module 9 to leave the EAT loop open. The EL MOT UP input voltage will, therefore, be zero. The absolute value of the EL MOT UP input, after scaling, shall be less than 0.079 Vdc, otherwise F131 shall be output. Run TPS module 10 to combine an azimuth yoke command of 0 Vdc with a Sight Rate Azimuth Command of +10 Vdc, output the difference from the DAC, and close the EAT loop through the IDE. The scaled EL MOT UP voltage shall then be 1.42 ± 0.079 Vdc, otherwise F131 shall be output.

4.7.3.8.2.7 EL MOT DOWN. The FCP TPS connects the EL MOT DOWN input at J8-FF, J8-HH (RTN) to the output of the Sight Command El DAC through Interface Device "E", when the TPS commands the external EAT loop to close. Run TPS module 9 to leave the EAT loop open. The EL MOT DOWN input voltage will, therefore, be zero. The absolute value of the EL MOT DOWN input, after scaling, shall be less than 0.079 Vdc, otherwise F132 shall be output. Run TPS module 10 to combine an elevation yoke command of 0 Vdc with a Sight Rate Elevation Command of +10 Vdc, output the difference from the DAC, and close the EAT loop through the IDE. The scaled EL MOT DOWN voltage shall then be 1.42 ± 0.079 Vdc, otherwise F132 shall be output.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

4.7.3.9 Analog output signals.

4.7.3.9.1 GUN COMMAND EL output. Run TPS module 9 to call subsequence 4 for setting the Gun Rate Elevation Command (GREC) to 5.0003968 Vdc, using BIT Mode 2 for SCB DAC to SCB ADC internal EAT. The Gun Command El DAC outputs this voltage to J8-(F) (HI) and J8-(G) (LO) with the limit relays de-energized and no EL TACH input. The ADC's input amplifier for summing all DAC outputs inverts and dc offsets this voltage (with all other amplifier inputs set to zero). The ADC measured value minus the dc offset shall equal +5.0003968 ± 0.1269632 Vdc, otherwise F031 shall be output.

Subsequence 4 then sets GREC to 5.0007020 Vdc and obtains the ADC measured value. This measured value is subtracted by software from the previously measured value and the result is compared to 0 ± 0.0097664 Vdc. If the result is out of tolerance, F031 shall be output.

Subsequence 4 also sets GREC to +5.0003968 Vdc and obtains the ADC measured value. The ADC measured value minus the dc offset shall equal -5.0003968 ± 0.1269632 Vdc, otherwise F031 shall be output.

Subsequence 4 then sets GREC to +5.0007020 Vdc and obtains the ADC measured value. This measured value is subtracted by software from the previously measured value and the result is compared to 0 ± 0.0097664 Vdc. If the result is out of tolerance, F031 shall be output.

Run TPS module 9 to set the Gun Rate Elevation Command (GREC) to +1.22 radians/second with the yoke inputs ignored (BIT Mode 3). The SCP's Gun Elevation Command (SGEC), using GREC in its re-iterative algorithm, will saturate at 1.5 Vdc. The Gun Command El DAC shall output this voltage, and with the DOWN LIMIT relay energized and no EL TACH input, a diode shall drop this voltage approximately 0.5 Vdc, leaving approximately 1.0 Vdc applied to the GUN COMMAND EL output at J8-(F) (HI) and J8-(G) (LO). The GUN COMMAND EL output voltage, as measured by the IOB shall be between 0.7 and 1.3 Vdc, otherwise F125 shall be output.

TPS module 9 also sets GREC to -1.22 radians/second with the yoke inputs ignored (BIT Mode 3). SGEC will saturate at +1.5 Vdc. The Gun Command El DAC shall output this voltage, but with the DOWN LIMIT relay energized, a diode shall block it. With no EL TACH input, the absolute value of the voltage at GUN COMMAND EL, as measured by the IOB shall be less than 0.079 Vdc, otherwise F125 shall be output.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

Run TPS module 10 to set GREC to -1.22 radians/second with the yoke inputs ignored (BIT Mode 3). SGEC will saturate at +1.5 Vdc. The Gun Command El DAC shall output this voltage and, with the UP LIMIT relay energized and no EL TACH input, a diode shall drop this voltage approximately 0.5 Vdc, leaving approximately 1.0 Vdc applied to the GUN COMMAND EL output. The GUN COMMAND EL output voltage, as measured by the IOB, shall be between +0.7 and +1.3 Vdc, otherwise F125 shall be output.

TPS module 10 also sets GREC to +1.22 radians/second with the yoke inputs ignored (BIT Mode 3). SGEC will saturate at -1.5 Vdc. The Gun Command El DAC shall output this voltage, but with the UP LIMIT relay energized, a diode shall block it. With no EL TACH input, the absolute value of the voltage at GUN COMMAND EL, as measured by the IOB, shall be less than 0.079 Vdc, otherwise F125 shall be output.

4.7.3.9.2 GUN COMMAND AZ 1 & GUN COMMAND AZ 2 outputs. Run TPS module 9 to call subsequence 4 for setting the Gun Rate Azimuth Command (GRAC) to -5.0003968 Vdc, using BIT Mode 2 for SCB DAC to SCB ADC internal EAT. The Gun Command Az DAC outputs this voltage to J8-H,N (HI) and J8-G,M (LO) with the limit relays de-energized and no AZ TACH input. The ADC's input amplifier for summing all DAC outputs inverts and dc offsets this voltage (with all other amplifier inputs set to zero). The ADC measured value minus the dc offset shall equal +5.0003968 ± 0.1269632 Vdc, otherwise F031 shall be output.

Subsequence 4 then sets GRAC to -5.0007020 Vdc and obtains the ADC measured value. This measured value is subtracted by software from the previously measured value and the result is compared to 0 ± 0.0097664 Vdc. If the result is out of tolerance, F031 shall be output.

Subsequence 4 also sets GRAC to +5.0003968 Vdc and obtains the ADC measured value. The ADC measured value minus the dc offset shall equal -5.0003968 ± 0.1269632 Vdc, otherwise F031 shall be output.

Subsequence 4 then sets GRAC to +5.0007020 Vdc and obtains the ADC measured value. This measured value is subtracted by software from the previously measured value and the result is compared to 0 ± 0.0097664 Vdc. If the result is out of tolerance, F031 shall be output.

Run TPS module 9 to set the Gun Rate Azimuth Command (GRAC) to +1.22 radians/second with the yoke inputs ignored (BIT Mode 3). The SCP's Turret Azimuth Command (STAC), using GRAC in its reiterative algorithm, will saturate at -3.0 Vdc. The Gun

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

Command Az DAC shall output this voltage, and with the CCW LIMIT relay energized and no AZ TACH input, a diode shall drop this voltage approximately 0.5 Vdc, leaving approximately -2.5 Vdc applied to the GUN COMMAND AZ 1 output at J8-H (HI), J8-G (LO) and the GUN COMMAND AZ 2 output at J8-N (HI), J8-M (LO). The GUN COMMAND AZ 1/AZ 2 output voltage, as measured by the IOB, shall be between -2.0 and -3.0 Vdc, otherwise F126 shall be output.

TPS module 9 also sets GRAC t -1.22 radians/second with the yoke inputs ignored (BIT Mode 3). STAC will saturate at +3.0 Vdc. The Gun Command Az DAC shall output this voltage, but with the CCW LIMIT relay energized, a diode shall block it. With no AZ TACH input, the absolute value of the voltage at GUN COMMAND AZ 1/AZ 2, as measured by the IOB, shall be less than 0.079 Vdc, otherwise F126 shall be output.

Run TPS module 10 to set GRAC to -1.22 radians/second with the yoke inputs ignored (BIT Mode 3). STAC will saturate at +3.0 Vdc. The Gun Command Az DAC shall output this voltage and, with the CW LIMIT relay energized and no AZ TACH input, a diode shall drop this voltage approximately 0.5 Vdc, leaving approximately +2.5 Vdc applied to the GUN COMMAND AZ 1 and GUN COMMAND AZ 2 outputs. The GUN COMMAND AZ 1/AZ 2 output voltage, as measured by the IOB, shall be between +2.0 and +3.0 Vdc, otherwise F126 shall be output.

TPS module 10 also sets GRAC to +1.22 radians/second with the yoke inputs ignored (BIT Mode 3). STAC will saturate at -3.0 Vdc. The Gun Command Az DAC shall output this voltage, but with the CW LIMIT relay energized, a diode shall block it. With no AZ TACH input, the absolute value of the voltage at GUN COMMAND AZ 1/AZ 2, as measured by the IOB, shall be less than 0.079 Vdc, otherwise F126 shall be output.

4.7.3.9.3 SIGHT COMMAND EL output. Run TPS module 9 to call subsequence 4 for setting the Sight Rate Elevation Command (SREC) to 5.0003968 Vdc, using BIT Mode 2 for SCB DAC to SCB ADC internal EAT. The Sight Command El DAC outputs this voltage to J4-(E) (HI) and J4-M (LO). The ADC's input amplifier for summing all DAC outputs inverts and dc offsets this voltage (with all other amplifier inputs set to zero). The ADC measured value minus the dc offset shall equal +5.0003968 ± 0.1269632 Vdc, otherwise F031 shall be output.

Subsequence 4 then sets SREC to -5.0007020 Vdc and obtains the ADC measured value. This measured value is subtracted by software from the previously measured value and the result is compared to 0 ± 0.0097664 Vdc. If the result is out of tolerance, F031 shall be output.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

Subsequence 4 also sets SREC to +5.0003968 Vdc and obtains the ADC measured value. The ADC measured value minus the dc offset shall equal -5.0003968 ± 0.1269632 Vdc, otherwise F031 shall be output.

Subsequence 4 then sets SREC to +5.007020 Vdc and obtains the ADC measured value. This measured value is subtracted by software from the previously measured value and the result is compared to 0 ± 0.0097664 Vdc. If the result is out of tolerance, F031 shall be output.

4.7.3.9.4 SIGHT COMMAND AZ output. Run TPS module 9 to call subsequence 4 for setting the Sight Rate Azimuth Command (SRAC) to -5.0003968 Vdc, using BIT Mode 2 for SCB DAC to SCB ADC internal EAT. The Sight Command Az DAC outputs this voltage to J4-(F) (HI) and J4-N (LO). The ADC's input amplifier for summing all DAC outputs inverts and dc offsets this voltage (with all other amplifier inputs set to zero). The ADC measured value minus the dc offset shall equal +5.0003968 ± 0.1269632 Vdc, otherwise F031 shall be output.

Subsequence 4 then sets SRAC to 5.0007020 Vdc and obtains the ADC measured value. This measured value is subtracted by software from the previously measured value and the result is compared to 0 ± 0.00097664 Vdc. If the result is out of tolerance, F031 shall be output.

Subsequence 4 also sets SRAC to +5.0003968 Vdc and obtains the ADC measured value. The ADC measured value minus the dc offset shall equal -5.0003968 ± 0.1269632 Vdc, otherwise F031 shall be output.

Subsequence 4 then sets SRAC to +5.0007020 Vdc and obtains the ADC measured value. This measured value is subtracted by software from the previously measured value and the result is compared to 0 ± 0.0097664 Vdc. If the result is out of tolerance, F031 shall be output.

Run TPS module 10 to set SRAC to 1 radian/second (+10 Vdc) in BIT Mode 2. The Sight Command Az DAC output at J4-(F) (HI), J4-N (LO) shall be coupled through Interface Device "E" to the AZ MOT 1 CW, AZ MOT 2 CW, and EL MOT UP inputs. These inputs shall have voltages, after scaling, of 1.42 ± 0.079 Vdc each, therefore, the SIGHT COMMAND AZ output voltage should be correct if these voltages are correct. If any motor voltage input is out of tolerance, F127, F129, or F131 shall be output, respectively.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the currentversion before use.

4.7.3.9.5 RANGE VOLTAGE output. Run TPS module 9 to output a range of zero meters for 1 ± 0.1 second, then increment the range in 20-meter steps at a rate of 2500 ± 250 meters/second up to 5000 meters (approximately 10 Vdc). At 5000 meters the range is reset to zero. If the range voltage exceeds approximately 8.5 Vdc, the TPS shall detect it as a logic high. If the range voltage output is detected logic high within 1.2 seconds, the test fails and a J4-(G) SA1 message shall be printed. If the range voltage output is detected logic low initially, then logic high within 3 seconds of logic low detection, the test passes. If the range voltage is not detected logic high within 4.2 seconds, fault message "J4-(G) SAO" shall be printed.

4.7.3.10 Gun command overrides.

4.7.3.10.1 DOWN LIMIT. Run TPS module 12 to input a -15.00 ± 0.75 Vdc GUN TACH EL voltage at J8-(X) (HI) and J8-(Y) (LO) and output +5 Vdc from the Gun Command El DAC. With +16 ± 1 Vdc applied to DOWN LIMIT at J8-Y and EL LIMIT COMMON at J8-Z, the DOWN LIMIT relay shall energize. Diodes shall block the +5 Vdc output from the DAC and allow the negative tachometer voltage to get through to the GUN COMMAND EL output at J8-(F) (HI) and J8-(G) (LO). The negative GUN COMMAND EL output voltage shall be measured by the IOB's ADC circuit, which doubles the voltage before applying it to the converter. This doubled voltage shall saturate the IOB's ADC at -5 Vdc. If the IOB's ADC is not negatively saturated, F125 shall be output.

Run TPS module 13 to input a +15.00 ± 0.75 Vdc GUN TACH EL voltage at J8-(X) (HI) and J8-Y (LO) and output +5 Vdc from the Gun Command El DAC. With +16 ± 1 Vdc applied to DOWN LIMIT at J8-Y and EL LIMIT COMMON at J8-Z, the DOWN LIMIT relay shall energize. Diodes shall block both the +5 Vdc output from the DAC and the positive tachometer voltage from being output at GUN COMMAND EL, J8-(F) (HI) and J8-(G) (LO). The absolute value of GUN COMMAND EL output voltage shall be less than 0.22 Vdc, otherwise F125 shall be output.

4.7.3.10.2 UP LIMIT. Run TPS module 11 to input a +15.00 ± 0.75 Vdc GUN TACH EL voltage at J8-(X) (HI) and J8-(Y) (LO) and output +5 Vdc from the Gun Command El DAC. With +16 ± 1 Vdc applied to UP LIMIT at J8-X and EL LIMIT COMMON at J8-Z, the UP LIMIT relay shall energize. Diodes shall block the -5 Vdc output from the DAC and allow the positive tachometer voltage to get through to the GUN COMMAND EL output at J8-(F) (HI) and J8-(G) (LO). The positive GUN COMMAND EL output voltage shall be measured by the IOB's ADC circuit, which doubles the voltage before applying it to the converter. This doubled voltage shall saturate the IOB's ADC at +4.96 Vdc. If the IOB's ADC is not positively saturated, F125 shall be output.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

Run TPS module 14 to input a -15.00 ± 0.75 Vdc GUN TACH EL voltage at J8-(X) (HI) and J8-(Y) (LO) and output -5 Vdc from the Gun Command El DAC. With +16 ± 1 Vdc applied to UP LIMIT at J8-X and EL LIMIT COMMON at J8-Z, the UP LIMIT relay shall energize. Diodes shall block both the -5 Vdc output from the DAC and the negative tachometer voltage from being output at GUN COMMAND EL, J8-(F) (HI) and J8-(G) (LO). The absolute value of GUN COMMAND EL output voltage shall be less than 0.22 Vdc, otherwise F125 shall be output.

4.7.3.10.3 CCW LIMIT. Run TPS module 12 to input a -15.00 ± 0.75 Vdc GUN TACH AZ voltage at J8-A (HI) and J8-B (LO) and output +5 Vdc from the Gun Command Az DAC. With +16 ± 1 Vdc applied to CCW LIMIT at J8-(B) and AZ LIMIT COMMON at J8-(C), the CCW LIMIT relay shall energize. Diodes shall block the +5 Vdc output from the DAC and allow the negative tachometer voltage to get through to the GUN COMMAND AZ 1 and GUN COMMAND AZ 2 outputs at J8-H (HI), J8-G (LO) and J8-N (HI), J8-M (LO), respectively. The negative GUN COMMAND AZ1/AZ 2 output voltage shall be measured by the IOB's ADC circuit. This voltage shall saturate the IOB's ADC at -5 Vdc. If the IOB's ADC is not negatively saturated, F126 shall be output.

Run TPS module 13 to input a +15.00 ± 0.75 Vdc GUN TACH AZ voltage at J8-A (HI) and J8-B (LO) and output +5 Vdc from the Gun Command Az DAC. With +16 ±1 Vdc applied to CCW LIMIT at J8-(B) and AZ LIMIT COMMON at J8-(C), the CCW LIMIT relay shall energize. Diodes shall block both the +5 Vdc output from the DAC and the positive tachometer voltage from being output at GUN COMMAND AZ 1 and GUN COMMAND AZ 2 at J8-H (HI), J8-G (LO) and J8-N (HI), J8-M (LO), respectively. The absolute value of GUN COMMAND AZ 1/AZ 2 output voltage shall be less than 0.2 Vdc, otherwise F126 shall be output.

4.7.3.10.4 CW LIMIT. Run TPS module 11 to input a +15.00 ± 0.75 Vdc GUN TACH AZ voltage at J8-A (HI) and J8-B (LO) and output +5 Vdc from the Gun Command Az DAC. With +16 ± 1 Vdc applied to CW LIMIT at J8-(A) and AZ LIMIT COMMON at J8-(C), the CW LIMIT relay shall energize. Diodes shall block the 5 Vdc output from the DAC and allow the positive tachometer voltage to get through to the GUN COMMAND AZ 1 and GUN COMMAND AZ 2 outputs at J8-H (HI), J8-G (LO) and J8-N (HI), J8-M (LO), respectively. The positive GUN COMMAND AZ 1/AZ 2 output voltage shall be measured by the IOB's ADC circuit. This voltage shall saturate the IOB's ADC at +4.96 Vdc. If the IOB's ADC is not positively saturated, F126 shall be output.

Run TPS module 14 to input a +15.00 ± 0.75 Vdc GUN TACH AZ voltage at J8-A (HI) and J8-B (LO) and output -5 Vdc from the Gun Command Az DAC. With +16 ± 1 Vdc applied to CW LIMIT at J8-(A)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the current version before use.

and AZ LIMIT COMMON at J8-(C), the CW LIMIT relay shall energize. Diodes shall block both the -5 Vdc output from the DAC and the negative tachometer voltage from being output at FUN COMMAND AZ 1 and GUN COMMAND AZ 2 at J8-H (HI), J8-G (LO) and J8-N (HI), J8-M (LO), respectively. The absolute value of GUN COMMAND AZ 1/AZ 2 output voltage shall be less than 0.2 Vdc, otherwise F126 shall be output.

4.7.3.11 Sight and gun synchro input signals.

4.7.3.11.1 Sight synchro inputs.

4.7.3.11.1.1 SIGHT SYNCHRO AZ. Run TPS module 9 to measure the sight position azimuth (SPA) output angle from the Synchro to Digital Converter (SDC) with no SIGHT SYNCHRO AZ input at J4-F (S1), G (S2), and H (S3) (external EAT loop through the IDE is open). SPA shall be 180 ± 0.0284 degrees, otherwise F702 shall be output.

Run TPS module 10 to close the EAT loop through the IDE, connecting the ANTENNA SYNCHRO AZ output at J4-X (S1), Y (S2), and Z (S3) to the SIGHT SYNCHRO AZ input at J4-F (S1), G (S2), and H (S3). SPA shall be accurate to within the values shown in Table XVII for the four Antenna Position Azimuth Command (APAC) words shown, otherwise F081 shall be output. Module 10 also checks the SDC's internal fault bit (SAB) to ensure it is logic low. If SAB is logic high, F123 shall be output.

TABLE XVII. Digital to synchro and synchro to digital external EAT conversion accuracy.

ANTENNA COMMAND APAC or APEC	EAT ACCURACY SPA, SPE, TPA, or GPE
4000H 90.006 degrees 1,570.7 milliradians	4000 ± C0H 90.006 ± 1.055 degrees 1,570.7 ± 18.4 milliradians
3FFFH 89.995 degrees 1,570.6 milliradians	3FFF ± C0H 89.995 ± 1.055 degrees 1,570.6 ± 18.4 milliradians
C000H 270.002 degrees 4,712.2 milliradians	C000 ± C0H 270.002 ± 1.055 degrees 4,712.2 ± 18.4 milliradians
BFFFH 269.996 degrees 4,712.1 milliradians	BFFF ± C0H 269.996 ± 1.055 degrees 4,712.1 ± 18.4 milliradians

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

4.7.3.11.1.2 SIGHT SYNCHRO EL. Run TPS module 9 to measure the sight position elevation (SPE) output angle from the SDC with no SIGHT SYNCHRO EL input at J4-(B) (S1), (C) (S2), and (D) (S3) (external EAT loop through the IDE is open). SPE shall be 180 ± 0.0284 degrees, otherwise F702 shall be output.

Run TPS module 10 to close the EAT loop through the IDE, connecting the ANTENNA SYNCHRO EL output at J4-A (S1), B (S2), and C (S3) to the SIGHT SYNCHRO EL input at J4-(B) (S1), (C) (S2), and (D) (S3). SPE shall be accurate to within the values shown in Table XVII for the four Antenna Position Elevation Command (APEC) words shown, otherwise F081 shall be output. Module 10 also checks the SDC's internal fault bit (SEB) to ensure it is logic low. If SEB is logic high, F123 shall be output.

4.7.3.11.1.3 GUN SYNCHRO AZ. Run TPS module 9 to measure the turret position azimuth (TPA) output angle from the SDC with no GUN SYNCHRO AZ input at J7-A (S1), B (S2), and C (S3) (external EAT loop through the IDE is open). TPA shall be 180 ± 0.0824 degrees, otherwise F702 shall be output.

Run TPS module 10 to close the EAT loop through the IDE, connecting the ANTENNA SYNCHRO AZ output at J4-X (S1), Y (S2), and Z (S3) to the GUN SYNCHRO AZ input at J7-A (S1), B (S2), and C (S3) (external EAT loop through the IDE is open). TPA shall be accurate to within the values shown in Table XVII for the four APAC words shown, otherwise F081 shall be output. Module 10 also checks the SDC's internal fault bit (TAB) to ensure it is logic low. If TAB is logic high, F123 shall be output.

4.7.3.11.1.4 GUN SYNCHRO EL. Run TPS module 9 to measure the gun position elevation (GPE) output angle from the SDC with no GUN SYNCHRO EL input at J7-D (S1), E (S2), and (A) (S3) (external EAT loop through the IDE is open). GPE shall be 180 ± 0.0284 degrees, otherwise F702 shall be output.

Run TPS module 10 to close the EAT loop through the IDE, connecting the ANTENNA SYNCHRO EL output at J4-A (S1), B (S2), and C (S3) to the GUN SYNCHRO EL input at J7-D (S1), E (S2), and (A) (S3). GPE shall be accurate to within the values shown in Table XVII for the four APEC words shown, otherwise F081 shall be output. Module 10 also checks the SDC's internal fault bit (GEB) to ensure it is logic low. If GEB is logic high, F123 shall be output.

4.7.3.12 Antenna synchro output commands. Run TPS module 9 to exercise software subsequence 9. In subsequence 9, the Self-Test Mode Control word (STMC) is set equal to 1 to do an internal

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

SIB EAT. This EAT connects the ANTENNA SYNCHRO AZ output to the SIGHT SYNCHRO AZ and EL inputs and connects the ANTENNA SYNCHRO EL output to the GUN SYNCHRO AZ and EL inputs through onboard relays. The four words output from the sight and gun SDC's are EAT Antenna Position Az Primary (EAPAP), EAT Antenna Position Az Redundant (EAPAR), EAT Antenna Position El Primary (EAPEP), and EAT Antenna Position El Redundant (EAPER). With the Antenna Az and El commands (APAC and APEC) set to the values listed in Table XVIII, the antenna position outputs shall have the accuracies shown in Table XVIII. If any output is out of tolerance, but both the SIB-to SCB bus and 115 VAC reference for the converters are good, F081 shall be output.

TABLE XVIII. Digital to synchro and synchro to digital internal EAT conversion accuracy.

ANTENNA COMMAND APAC or APEC	EAT ACCURACY EAPAP, EAPAR, EAPEP, or EAPER
2000H 45.000 degrees 785.4 milliradians	2000 ± C0H 45.000 ± 1.055 degrees 785.4 ± 18.4 milliradians
IFFFH 44.995 degrees 785.3 milliradians	1 FFF ± C0H 44.995 ± 1.055 degrees 785.3 ± 18.4 milliradians
E000H 315.002 degrees 5,497.6 milliradians	E000 ± C0H 315.002 ± 1.055 degrees 5,497.6 ± 18.4 milliradians
E001H 315.008 degrees 5,497.7 milliradians	E001 ± C0H 315.008 ± 1.055 degrees 5,497.7 ± 18.4 milliradians

4.7.4 Spare connector pins. Run TPS module 6 to pull J3-DD logic high. If J3-DD is stuck logic low, FCP power shall be turned off by the TPS. Connector-pins J3-(D), (K), (M), and J6-(J) are not tested by the TPS.

4.8 Soldering. Soldering processes shall be inspected by certified inspectors to the requirements and acceptance criteria of Note 2 of Drawing 12950877 as specified in paragraph 3.7.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the currentversion before use.

5. PACKAGING

5.1 General. Preservation, packing and marking shall be in accordance with Special Packaging Instruction PS 9360890 and MIL-P-14232. The level of protection shall be as specified in the procurement document.

6. NOTES

(This section contains information of a general or explanatory nature that may be helpful, but is not mandatory.)

6.1 Intended use. The fire control processor is used in the PIVADS self-propelled and towed vehicles.

6.2 Acquisition requirements. Acquisition documents must specify the following:

a. Title, number and date of this specification

b. Issue of DODISS to be cited in the solicitation, and, if required, the specific issue of individual documents referenced (see 2.1.1).

c. Requirements for submission of first article sample.

d. Applicable stock number.

e. Packaging requirements, if other than specified in Section 5.

f. Serialization requirements, if applicable.

g. Certificate of conformance for each lot or shipment of product.

6.3 Submission by the contractor.

a. Inspection equipment designs for approval. Submit two copies of designs as required to: Commander, U.S. Army ARDEC, ATTN: SMCAR-QAF-I, Picatinny Arsenal, NJ 07806-5000. This address will be specified on the Contract Data Requirements List, DD Form 1423 in the contract.

b. Alternative quality conformance provisions for approval. Unless otherwise specified in the contract, proposed alternative quality conformance provisions will be submitted by the contractor for evaluation to the technical activity responsible for the preparation of this specification.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the current version before use.

6.4 Drawings. Drawings listed in Section 2 of this specification under the heading U.S. Army Armament, Research, Development and Engineering Center (ARDEC) may also include drawings prepared by, and identified as U.S. Army Armament, Research and Development Command (ARRADCOM), Frankford Arsenal, Rock Island Arsenal or Picatinny Arsenal drawings. Technical data originally prepared by these activities is now under cognizance of ARDEC.

6.5 Subject term (key word) listing.

Built-In-Test (BIT)
Computer Memory Board (CMB)
Data Transmission Board (DTB)
End-Around-Test (EAT)
Input/Output Board (IOB)
Product Improved Vulcan Air Defense System (PIVADS)
Servo Controller Board (SCB)
Synchro Interface Board (SIB)

6.6 Reference material. Table XIX provides the conversion factors for the converters on the Input/Output Board (offset 8-bit ADC), Servo Controller Board (complementary-two's complement 12-bit ADC after inversion to two's complement by buffers U36 and U37, and offset 16-bit DAC), and on the Synchro Interface Board (16-bit synchro). Scaling factors for the various analog inputs, are provided by paragraphs 3.5.1.8.1 and 3.5.1.8.2. This information can be used to assist in failure analyses.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.

TABLE XIX. Conversion factors.

Bit (Active High)	Offset 8-bit (Vdc)	2's Complement 12-bit (Vdc)	Offset 16-bit (Vdc)	Synchro (deg)
D15 (MSB)	--	Sign (Neg)	Sign (Pos)	180.0
D14	--	5.000	5.000	90.00
D13	--	2.500	2.500	45.00
D12	--	1.250	1.250	22.50
D11	--	0.625	0.625	11.25
D10	--	0.3125	0.3125	5.625
D9	--	0.15625	0.15625	2.8125
D8	--	0.07812	0.07812	1.40625
D7 (MSB)	Sign (Pos)	0.03906	0.03906	0.70312
D6	2.50	0.01953	0.01953	0.35156
D5	1.25	0.009765	0.009765	0.17578
D4	0.625	0.004883	0.004883	0.08789
D3	0.312	--	0.002441	0.04395
D2	0.156	--	0.001221	0.02197
D1	0.078	--	0.0006101	0.01099
D0 (LSB)	0.039	--	0.0003052	0.005493

NOTES:

1. Offset 8-bit data is transferred as one 8-bit byte, and 12-bit data is transferred as one 16-bit word with the least significant bits set to a logic "0", after inversion by SCB buffer U37.

2. Hexadecimal 80 (8-bit), 000 (12-bit) and 8000 (16-bit) correspond to a converted voltage of 0.00 Vdc.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:29Z
Check the source to verify that this is the current version before use.

6.7 Changes from previous issue. The previous issues were for ARDEC internal use only. They were not forwarded to Naval Publications and Forms Center (NPFC) for publication. This issue is initial public release for unlimited distribution.

Custodian:
Army-AR

Preparing activity:
Army-AR

(Project 1285-A648)

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:29Z
Check the source to verify that this is the current version before use.